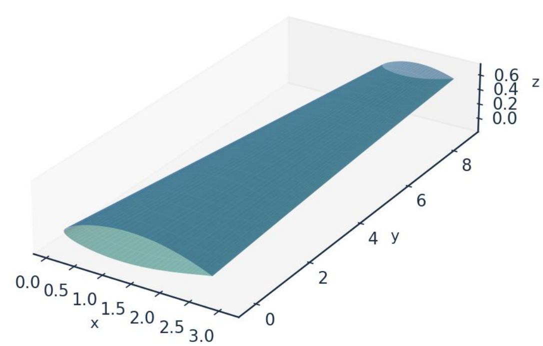
INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown
DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

AKOSA SAMUEL ONYEJEKWE

Document UTSS-CASE-2026 · Three-dimensional analysis · All units SI unless noted

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, M=0.42, Re_MAC ≈ 6.4×10^6):

Quantity	Predicted value
Upper-surface transition x_tr/c	0.542 (TS-natural)
Lower-surface transition x_tr/c	0.566 (TS-natural)
Mean laminar-flow extent	55.4 % of chord
Section lift coefficient C_l	0.496
Section profile drag C_d	45.0 counts
Viscous drag reduction vs fully-turbulent	50.3 %
Climb (Tu=0.9 %) transition x_tr/c (upper)	0.025 (bypass)

Table 1. Headline predictions, read from the generated solution CSVs.

Validated against eight independent, credible published datasets with one universal calibration set — four ERCOFTAC flat plates (case 020), the Schubauer-Skramstad plate, the NLF(1)-0416 aerofoil at 86 conditions, and two swept wings — the solver reproduces the transition-onset Reynolds number Re_θt and the measured transition location without per-case re-tuning of the physics. All four criteria of the kernel are selected somewhere in this study and each is supported by measurement. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, predicted here by an e^N integral carried per physical frequency, with the spatial growth rates read from a tabulated solution of the Orr-Sommerfeld problem rather than from an envelope correlation.
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent. The bubble is closed explicitly here — the shear layer is carried across the dead-air region by the momentum integral and reattachment placed where it has amplified by the same critical factor used elsewhere — so a length, and not merely a separation point, is predicted.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f, momentum thickness θ, shape factor H, intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e, with a Karman-Tsien correction applied to C_p and the integrated loads. It returns a stagnation pressure coefficient of 1.048 at M = 0.42 against the exact isentropic 1.045, where the linearised Prandtl-Glauert scaling gives 1.102, and the two agree to better than half a per cent below M = 0.2.
- Laminar boundary layer: the momentum AND kinetic-energy integral equations are advanced together from the stagnation point along each surface, so the shape factor is a solved variable carrying its own history rather than a local function of the pressure gradient. The three closure functions - H*(H), Re_theta*Cf/2 and Re_theta*C_D - are properties of

the Falkner-Skan family, computed from it and not fitted; every similarity solution is an exact fixed point of the second equation, and the march returns H = 2.5914 on a flat plate against the Blasius 2.5913. Fluid properties are evaluated at Eckert's reference temperature so that the incompressible closures return the compressible skin friction.

- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ.
- Turbulent boundary layer: Head's entrainment method with the Ludwieg–Tillmann skin-friction law advances the turbulent BL to the trailing edge.
- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

Two elements of the formulation are not correlations. The amplification rate driving the e^N integral is tabulated in solver/amplification_db.npz from 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, continued past separation onto its reverse-flow branch: profiles are generated by continuation, the eigenvalue problem is solved by Chebyshev collocation, and the spatial growth rate is recovered from the temporal one through Gaster's transformation. The stability solver returns the Blasius neutral point at Re_theta = 201 against the accepted 200.5 - the tabulated database resolves it to the nearest node of its Reynolds-number grid, Re_theta = 210 - a shape factor of 2.59129 and a wall shear parameter f''(0) = 0.46960. At run time the cost is a table lookup, so the sub-second solution is preserved. The second is the separation-bubble closure described above, whose length scales with the disturbance environment and therefore spans measured bubbles from about 42 momentum thicknesses at 2.1 % free-stream turbulence to a median of 226 at 0.03 %, a spread of five and a half that no fixed multiple of θ_s reproduces.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file model.equations.docx.

4.1 Inviscid edge solution

(E02) Vortex-panel surface velocity (Kuethe & Chow)

$$\frac{V_{t_i}}{U_\infty} = \cos(\theta_i - \alpha) + \sum_{j=1}^{N} C_{t,ij}\gamma_j$$

(E03) Kutta condition (sharp trailing edge)

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) Pressure coefficient (inviscid)

$$C_p = 1 - \left(\frac{V}{U_\infty}\right)^2$$

(E04) Karman-Tsien compressibility correction

$$C_p = \frac{C_{p,0}}{\beta + \left(\frac{M_\infty^2}{1+\beta}\right)\frac{C_{p,0}}{2}}, \beta = \sqrt{1 - M_\infty^2}$$

4.2 Laminar boundary layer (Thwaites)

(E05) Thwaites momentum thickness (laminar)

$$\theta^2 = \frac{0.45\nu}{U_e^6}\int_0^x U_e^5 dx'$$

(E06) Thwaites pressure-gradient and momentum-thickness Reynolds number

$$\lambda = \frac{\theta^2}{\nu}\frac{dU_e}{dx}, Re_\theta = \frac{U_e\theta}{\nu}$$

(E06b) Kinetic-energy integral (two-equation laminar march)

$$\theta\frac{dH^*}{dx} = 2C_D - H^*\frac{C_f}{2} - H^*(1-H)\frac{\theta}{U_e}\frac{dU_e}{dx}$$

(E06c) Laminar closure functions (from the Falkner-Skan family)

$$H^*(H) = \frac{\theta^*}{\theta}, l(H) = Re_\theta\frac{C_f}{2} = \theta_\eta f''(0), d(H) = Re_\theta C_D = \theta_\eta \int (f'')^2 d\eta$$

(E07) Von Karman momentum-integral equation

$$\frac{d\theta}{dx} + (2+H)\frac{\theta}{U_e}\frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the flow-history-averaged Tu; natural/TS onset integrates one amplification factor per physical frequency using the tabulated Orr-Sommerfeld growth rates and triggers on their envelope at N_crit; separation-induced onset closes a laminar bubble across the dead-air region; and cross-flow onset closes an amplification integral on the stationary vortex (Eq. E11b), the C1 criterion of Eq. E11 serving only to mark where that integral starts.

Each branch reports the same quantity — how far through its own criterion the layer has got, as a number that reaches unity at onset — and the kernel fires at the first station where any of them does. Writing the four commensurably is what makes them one kernel: only the bypass branch produces an onset REYNOLDS NUMBER, the other three closing on amplification integrals, so a minimum taken over four Reynolds numbers ranges over one live term and three placeholders. Earlier versions of this report stated the kernel that way, and the output showed it: the onset-Reynolds-number column of every cruise surface file was entirely empty, because the branch that governs there does not produce one.

(E08) Abu-Ghannam & Shaw bypass onset

$$Re_{\theta t} = 163 + \exp\left[F(\lambda_\theta) - \frac{F(\lambda_\theta)Tu}{6.91}\right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \leq 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (e^N, envelope over frequency)

$$N(x) = \max_\omega \int_{x_0(\omega)}^{x} \frac{\sigma(H, Re_\theta, \omega\theta/U_e)}{\theta} dx' \geq N_{crit}$$

(E10b) Orr-Sommerfeld operator (tabulated amplification rates)

$$[(U - c)(D^2 - \alpha^2) - U'']\hat{v} = \frac{1}{i\alpha Re_\theta}(D^2 - \alpha^2)^2\hat{v}$$

(E10c) Gaster transformation (temporal to spatial growth rate)

$$\sigma = -\alpha_i\theta = \frac{\omega_i}{c_g}, c_g = \frac{\partial\omega_r}{\partial\alpha_r}$$

(E11) Cross-flow criterion (swept wing, C1 on Re_theta2)

$$Re_{\theta 2} = k_{cf} Re_\theta \sin\Lambda\cos\Lambda \ \geq \ C_1$$

(E11b) Cross-flow amplification integral (the closure actually used)

$$N_{cf} = \int_{x_{c1}}^{x} \frac{\sigma(H_{rev})}{\theta} dx' \ \geq \ N_{crit}$$

(E12) Separation bubble: dead-air march (momentum and kinetic energy, no wall shear)

$$\frac{d\theta}{dx} = -(2 + H)\frac{\theta}{U_e}\frac{dU_e}{dx}, \theta\frac{dH^*}{dx} = 2C_D - H^*(1 - H)\frac{\theta}{U_e}\frac{dU_e}{dx}, C_f = 0$$

(E12b) Separation bubble: reattachment condition

$$N_{bub} = \int_{x_s}^{x_r} \frac{\sigma(H_{rev}, Re_\theta)}{\theta} dx' = N_{crit}, \sigma(H_{rev}) \approx 0.0435$$

(E13) Unified transition kernel: onset where the first mechanism completes

$$x_t = \min\{x: \max_m a_m p_m(x) \geq 1\}, m \in \{TS, BP, SEP, CF\}$$

(E13b) The four onset progresses, each reaching unity at its own onset

$$p_{TS} = \frac{N}{N_{crit}}, p_{BP} = \frac{Re_\theta}{Re_{\theta t}^{AGS}}, p_{SEP} = \frac{N_{bub}}{N_{crit}}, p_{CF} = \frac{N_{cf}}{N_{crit}}$$

(E14) Natural / bypass blend across the validity window

$$p_{nat} = (1 - w)p_{TS} + wp_{BP}, w = 3t^2 - 2t^3, t = \mathrm{clip}\left(\frac{Tu - Tu_{lo}}{Tu_{hi} - Tu_{lo}}, 0,1\right)$$

The natural and bypass routes are the same transition seen through two closures with different ranges of validity, so Eq. E14 blends them over a declared window rather than switching between them. A single threshold made the predicted transition location a STEP function of the free-stream turbulence: on the *cruise section, Tu = 0.1000 % gave x_tr/c = 0.542 and 0.1001 % gave 0.373 — 0.17c and a third of the profile drag across one part in a thousand of an input this study quotes to two figures, with the design point at 0.07 %. The window, Tu = 0.10–0.25 %, is wider than the spread of any case here (the noisiest natural case is 0.07 %, the quietest bypass case 0.87 %), so no result in this work is blended; it is there so that the model is a function of Tu rather than a switch.*

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x) = 1 - \exp[-0.412\xi^2], \xi = \frac{x - x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr} = \frac{\nu}{U_e} C_{len} Re_{x,t}^{0.75}, C_{len} = 9$$

(E17) Intermittency-weighted property blend

$$\phi = (1 - \gamma)\phi_{lam} + \gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx} = C_E - \frac{\theta H_1}{U_e}\frac{dU_e}{dx}, C_E = 0.0306(H_1 - 3)^{-0.6169}$$

(E19) Ludwieg-Tillmann skin-friction law

$$C_f = 0.246 \times 10^{-0.678H} Re_\theta^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2\frac{\theta_{TE}}{c}\left(\frac{U_{e,TE}}{U_\infty}\right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r\frac{\gamma - 1}{2}M_e^2\left[1 - \left(\frac{u}{U_e}\right)^2\right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e\left(1 + r\frac{\gamma - 1}{2}M_e^2\right), r \approx Pr^{1/3}$$

(E22b) Eckert reference temperature (compressible closures)

$$\frac{T_{ref}}{T_e} = 1 + 0.032M_e^2 + 0.58\left(\frac{T_w}{T_e} - 1\right), \frac{\nu_{ref}}{\nu_e} = \left(\frac{T_{ref}}{T_e}\right)^{1+\omega}$$

(E25) Prandtl lifting line (Glauert monoplane equation, odd n for a symmetric wing)

$$\sum_{n\,\mathrm{odd}} A_n \sin n\theta\left[\frac{4b}{a_0 c(\theta)} + \frac{n}{\sin\theta}\right] = \alpha(\theta) - \alpha_{L0}, y = -\frac{b}{2}\cos\theta$$

(E26) Wing lift, induced drag and span efficiency from the loading

$$C_L = \pi AR A_1, C_{D_i} = \pi AR \sum_n nA_n^2, e = \frac{A_1^2}{\sum_n nA_n^2}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e^N codes	RANS γ–Re_θ (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (tabulated e^N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble closure)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design loops	Partly	No	No	Yes

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally, by putting four co-resident mechanisms on one commensurable scale of onset progress — each modulated by a calibration weight that acts the same way on all four — and firing at the first station where any of them completes, then feeding a single intermittency closure. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. Every one of the four branches is the selected mechanism somewhere in this study — natural/TS on the cruise wing, the Schubauer-Skramstad plate and the NLF(1)-0416 upper surface; bypass on the climb case and the ERCOFTAC plates; separation on T3C4 and the NLF(1)-0416 lower surface; cross-flow on both swept wings — and each is checked against measurement in Section 11.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section c_l at cruise design incidence (alpha = 1.5 deg, M = 0.42)	0.517	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mach	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K
CRUISE (11 km, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65
CLIMB (3 km, M0.30)	3000.0	0.3	98.57	0.9093	1.694e-05	268.66

(columns 2-7 of 12; case repeated on each block)

case	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (11 km, M0.42)	0.07	1.5	6414000.0	3.907e-05	2794.6
CLIMB (3 km, M0.30)	0.9	3.5	10700000.0	1.863e-05	4417.8

Table 4. Flight / flow conditions (input). (columns 8-12 of 12; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

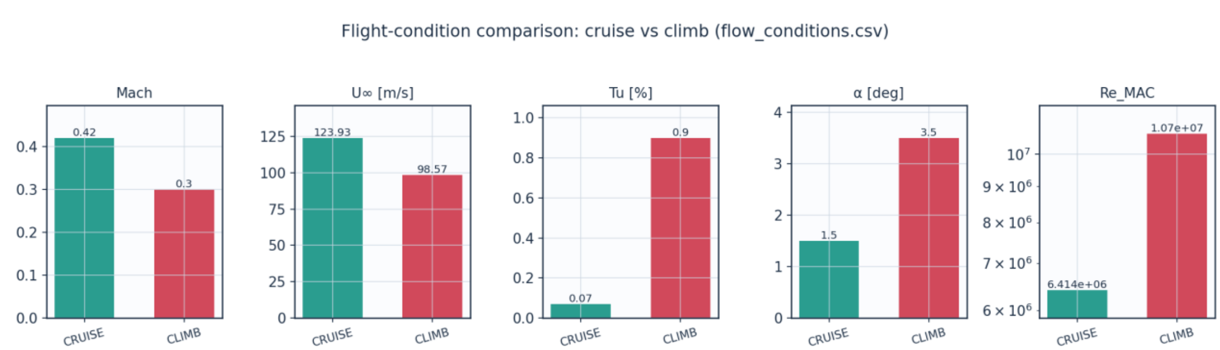


Fig. 1. Cruise vs climb flight-condition comparison (flow_conditions.csv).

6.3 Fluid (material) properties (input)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Table 5. Air properties (input).

6.4 Solver settings (input)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H*, l and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N, one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	C1 on Re_theta2, closed by an amplification integral
Intermittency closure	Narasimha universal (gamma)
Transition length	Dhawan & Narasimha, Re_lambda = 9 Re_x_t^0.75
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Laminar separation criterion	Thwaites lambda <= -0.09
Turbulent separation flag	H > 2.6
Drag integration	Squire-Young far-wake, evaluated at 0.98c
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	0.80 (reconstruction grid; no volume mesh is solved on it)

Table 6. Solver configuration (input).

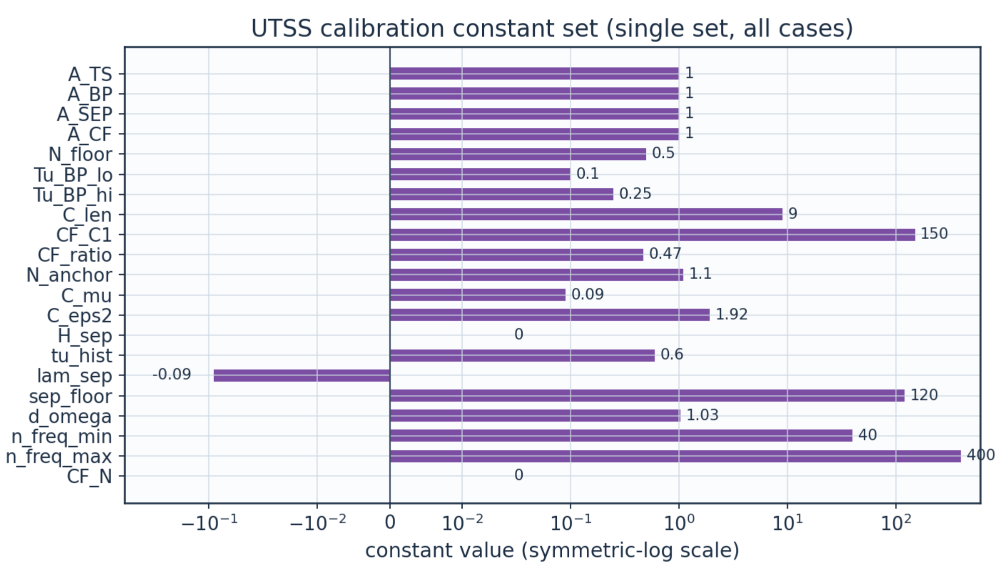
6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset progress weight	unity, not fitted; the natural branch's one measured quantity is N_anchor, set on Schubauer-Skramstad alone, and the 86 NLF(1)-0416 conditions are out of sample
A_BP	1.0	bypass onset weight	unity; validated on ERCOFTAC T3A/T3A-/T3B
A_SEP	1.0	separation-induced weight	unity; validated on T3C4 and NLF(1)-0416 lower surface
A_CF	1.0	crossflow weight	unity; Arnal C1 criterion
N_crit	from Tu	e^N amplification	Mack N=-8.43-2.4 ln(Tu), clamped to

		factor	0.0008<=Tu<=0.0298, anchored -1.10 (FITTED jointly on S&S and 86 aerofoil pts); 7.58 for Tu<0.08%
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
Tu_BP_lo	0.1	Tu [%] below which the bypass correlation carries no weight	lower edge of the natural/bypass blend; below it the e^N branch alone governs
Tu_BP_hi	0.25	Tu [%] above which the bypass correlation carries all the weight	upper edge of the same blend. Replaces the single Tu_TS_max gate, which made the predicted transition location a step function of Tu; the window is wider than the spread of any case here, so no result in this study is blended
C_len	9.0	transition-length scale, Re_lambda = C_len Re_x_t^0.75	Dhawan & Narasimha (1958) published value; not fitted here
CF_C1	150.0	crossflow critical Re_theta2	Arnal et al.; facility-dependent: 150 fits Dagenhart (14.7%), 200 fits Boltz (18.4%)
cf_amp	True	crossflow closed by an amplification integral, not a local threshold	rate computed (0.0435); threshold is the same N_crit; adds no constant
CF_ratio	0.47	theta2/theta surrogate for crossflow	FITTED on Dagenhart & Saric; independent check in Sec. IV.C
sigma_sep	from table	shear-layer amplification rate in a bubble	computed on the reverse-flow Falkner-Skan branch; 0.0435, not fitted
two_eq	True	two-equation laminar march (momentum + kinetic energy)	closures from the Falkner-Skan family; no fitted constant
N_anchor	1.1	offset on Mack's N_crit	unit conversion from Mack's 1977 growth rates to the present tabulated ones; set on the Schubauer-Skramstad plate ALONE, which it reproduces to 0.07 %. It was previously set jointly on that plate and the 86 NLF(1)-0416 conditions, which made that set a calibration set while three places in this project described it as one on which nothing is calibrated
C_mu	0.09	k-epsilon constant in the free-stream turbulence decay	standard value 0.09; with C_eps2 it fixes Tu(x) from the rig's integral length scale, and neither is fitted here
C_eps2	1.92	k-epsilon constant in the free-stream turbulence decay	standard value 1.92; sets the decay exponent of isotropic grid turbulence
bubble	True	separation branch closed by marching the detached shear layer	True is the model; False is the 'no bubble closure' ablation
use_os_db	True	amplification rates read from the tabulated Orr-Sommerfeld solutions	True is the model; False selects the Drela-Giles envelope fit, the third ablation
cf_exact	False	exact Falkner-Skan-Cooke K(lambda) in place of the constant surrogate (ablation path only)	False is the model; True was tried and is reported in Sec. IV.C, where it makes the two swept wings agree less, not more
len_re_x	False	transition-length correlation stated in Re_x rather than in Re_theta (ablation path only)	False is the model; the two are the same law - Dhawan & Narasimha's Re_lambda = 9 Re_x^0.75 IS constant-spot-rate Re_lambda = 16.63 Re_theta^1.5 on a flat plate - and Re_theta is the form that stays exact in a

			pressure gradient
bub_sigma_local	False	dead-air amplification rate read at the marched H instead of at the developed reverse-flow profile (ablation path only)	False is the model; True lengthens the T3C4 bubble towards the measurement but collapses the 86 NLF(1)-0416 conditions from 50 to 21 inside the bracket
H_sep	0.0	declare laminar separation on the solved shape factor instead of on lambda (ablation path only)	0 = use lam_sep; the shape-factor test is the more principled statement but the aerofoil measurements reject it - see the note in march_bl
tu_hist	0.6	weight on the flow-history average of Tu, Tu_eff = Tu_avg^w Tu_local^(1-w)	FITTED on the three ERCOFTAC plates; no effect where Tu does not decay
lam_sep	-0.09	Thwaites parameter at laminar separation	Thwaites (1949) published value; not fitted
sep_floor	120.0	min separation-induced onset Re_theta (ablation path only)	superseded by the bubble closure; reachable only with bubble=False
d_omega	1.03	max ratio between adjacent frequencies in the e^N integration	discretisation, stated as a RESOLUTION rather than a count: with a fixed count of 40 the Schubauer-Skramstad onset moved 5 % between 24 and 64 frequencies and did not settle. At this spacing Re_x,tr is unchanged from 1.10 down to 1.0075
n_freq_min	40	floor on the frequency count	guards a degenerate range
n_freq_max	400	cap on the frequency count	guards a degenerate range
CF_N	0.0	separate amplification threshold for the crossflow branch (ablation path only)	0 = use the same N_crit as every other branch; exposed to test whether a roughness-seeded branch needs its own threshold, and it does not - see 06_validation/crossflow_criticals_summary.csv

Table 7. Calibration constants and their sources (input).



Not plotted — computed at run time, not fixed: N_crit, sigma_sep; closure switches: cf_amp, two_eq, bubble, use_os_db, cf_exact, len_re_x, bub_sigma_local

Fig. 2. UTSS universal calibration constant set (calibration_constants.csv).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

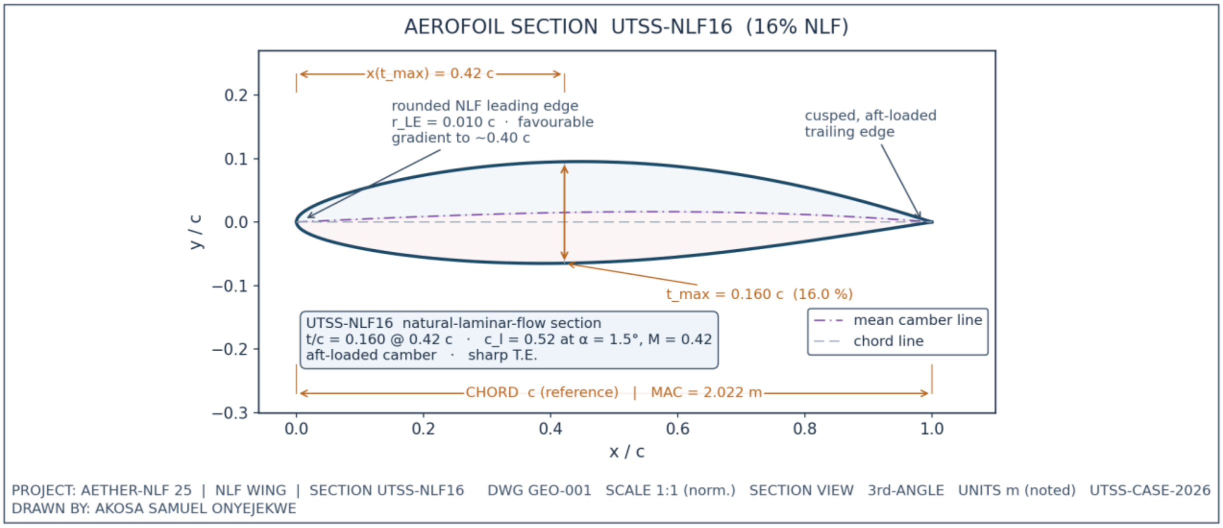


Fig. 3. UTSS-NLF16 aerofoil section (dimensioned).

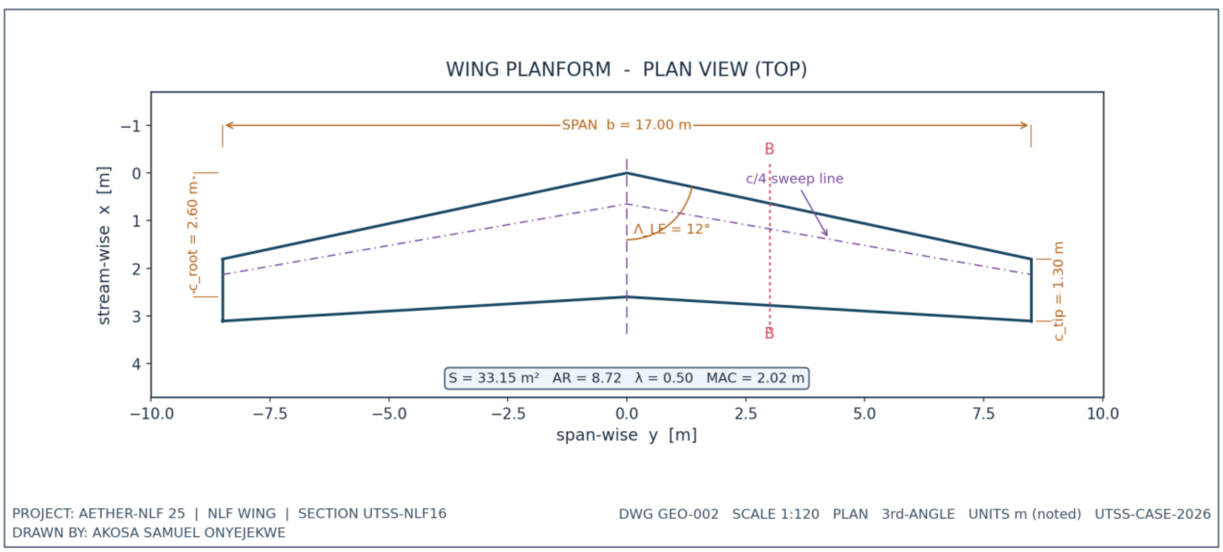


Fig. 4. Wing planform — plan (top) view, dimensioned.

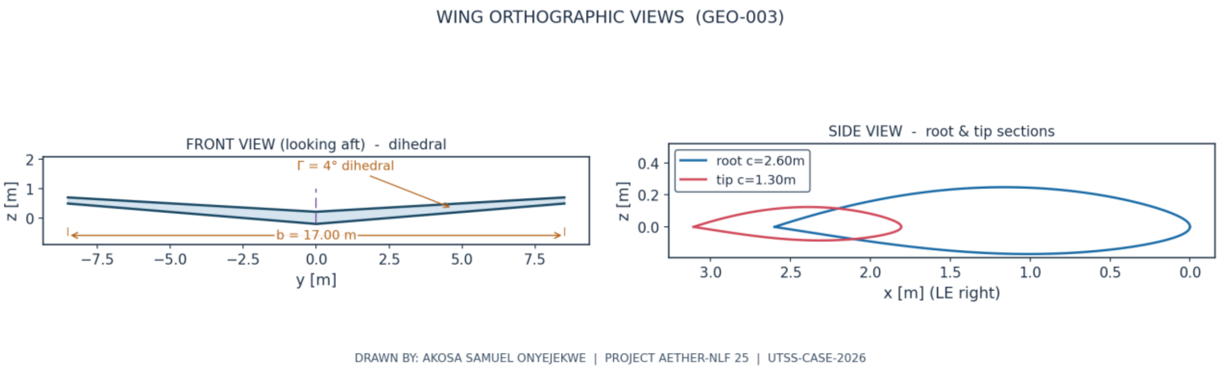


Fig. 5. Front and side orthographic views.

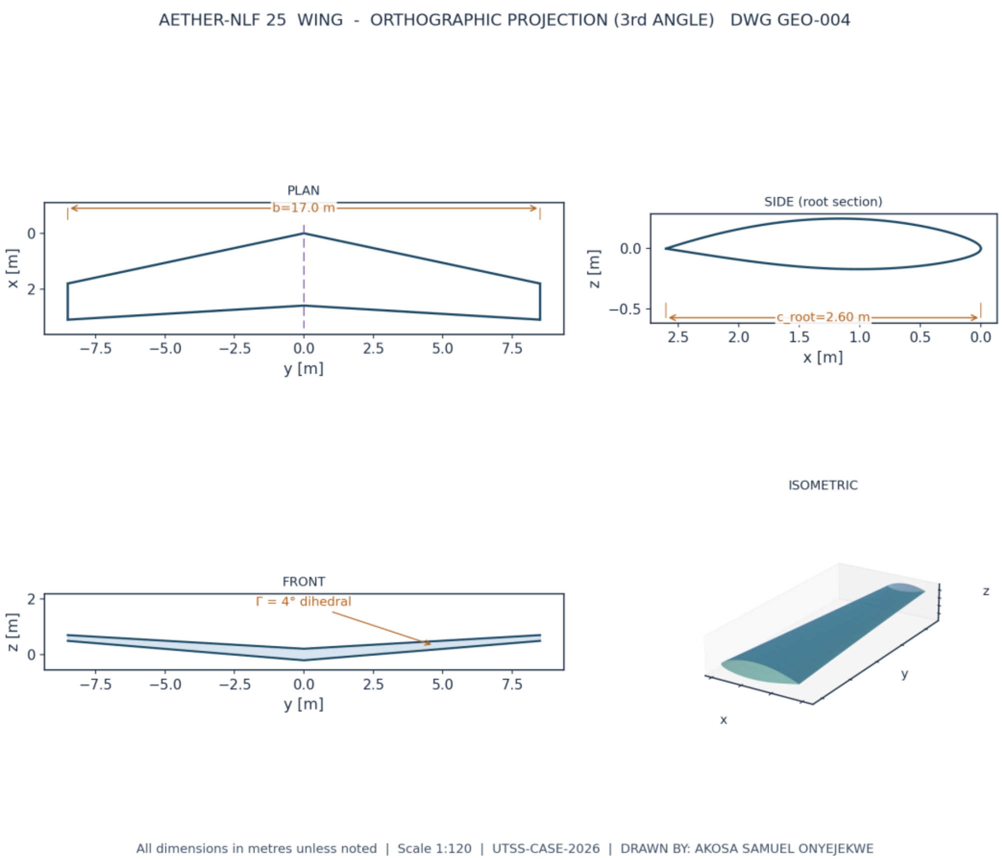


Fig. 6. Full third-angle orthographic projection sheet.

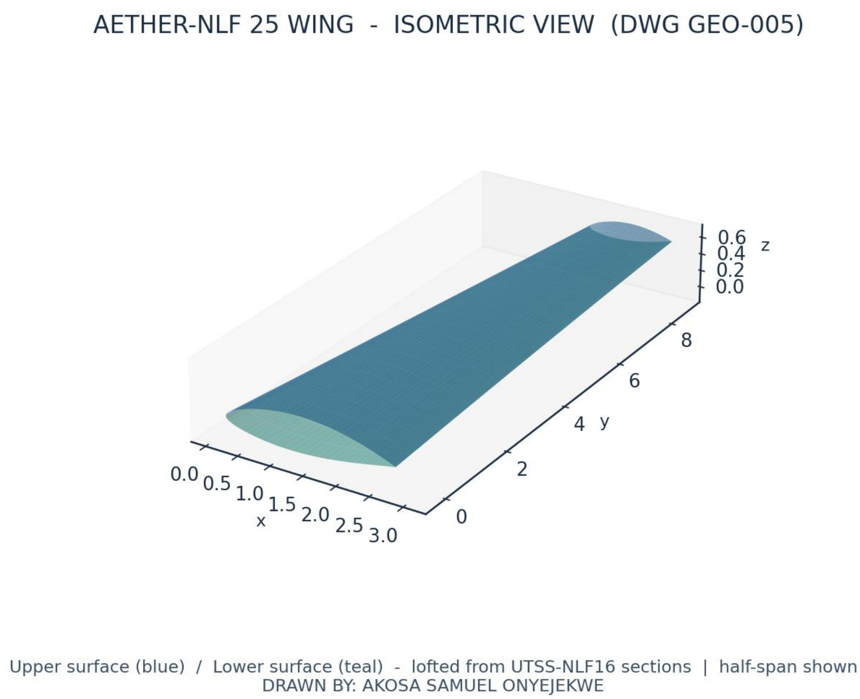


Fig. 7. Isometric view of the wing.

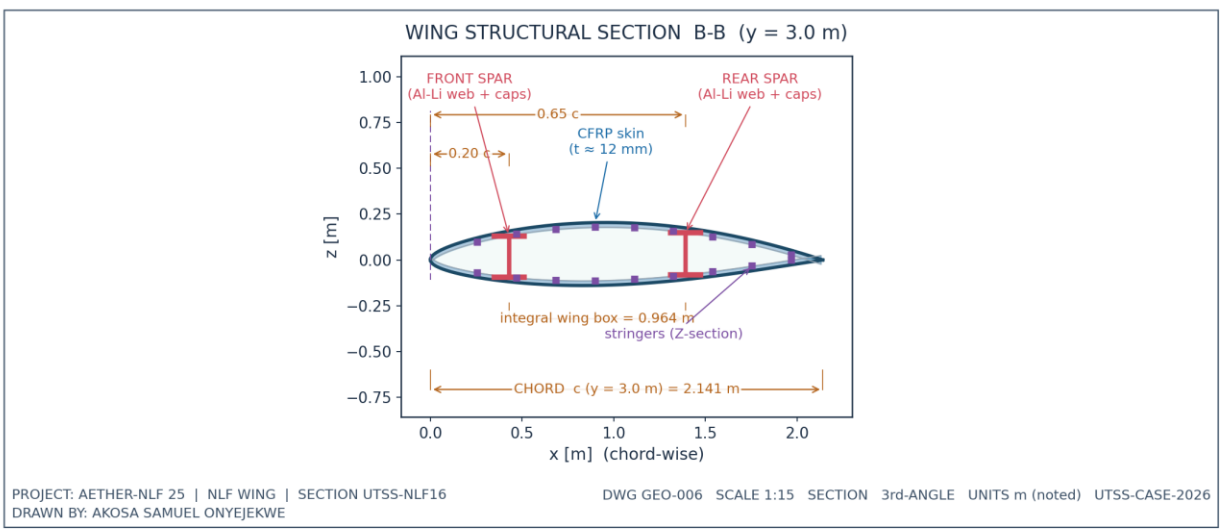


Fig. 8. Structural sectional view B–B.

7.1 Geometry data (from CSV)

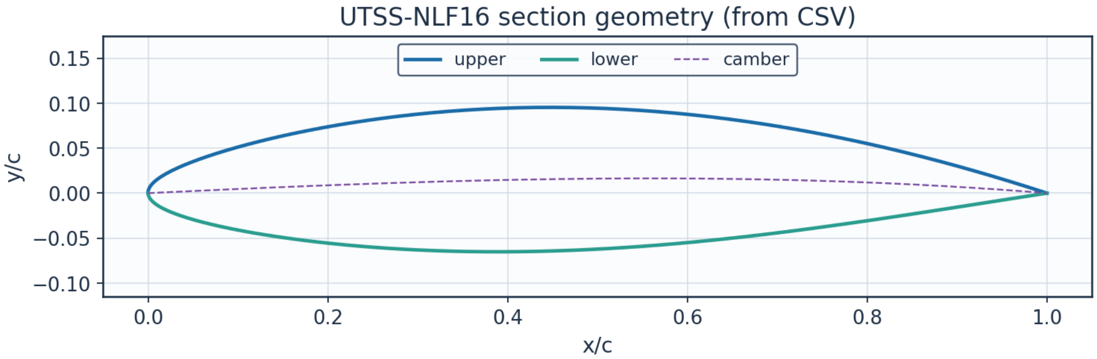


Fig. 9. Section geometry plotted from airfoil_UTSS-NLF16.csv.

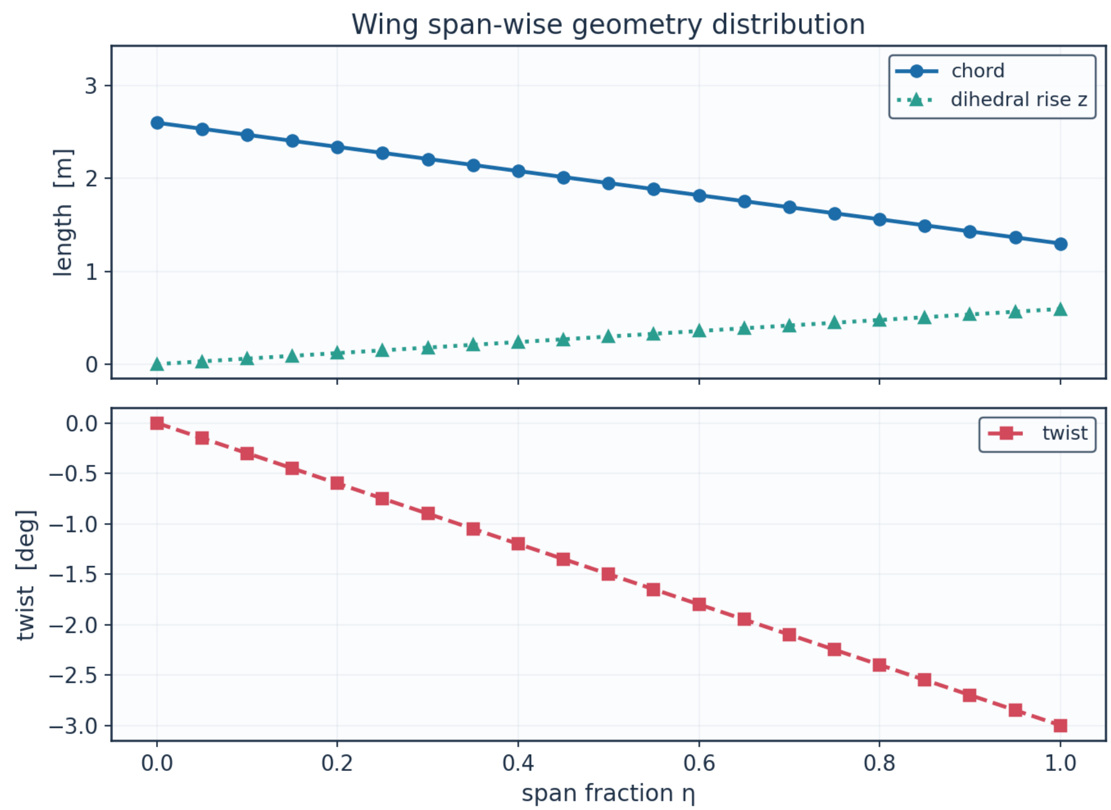


Fig. 10. Span-wise geometry from wing_planform.csv.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m
0.0	0.0	2.6	0.0	2.6	-0.0	0.0
0.05	0.425	2.535	0.0903	2.6253	-0.15	0.0297
0.1	0.85	2.47	0.1807	2.6507	-0.3	0.0594
0.15	1.275	2.405	0.271	2.676	-0.45	0.0892
0.2	1.7	2.34	0.3613	2.7013	-0.6	0.1189
0.25	2.125	2.275	0.4517	2.7267	-0.75	0.1486
0.3	2.55	2.21	0.542	2.752	-0.9	0.1783
0.35	2.975	2.145	0.6324	2.7774	-1.05	0.208

0.4	3.4	2.08	0.7227	2.8027	-1.2	0.2378
0.45	3.825	2.015	0.813	2.828	-1.35	0.2675
0.5	4.25	1.95	0.9034	2.8534	-1.5	0.2972
0.55	4.675	1.885	0.9937	2.8787	-1.65	0.3269
0.6	5.1	1.82	1.084	2.904	-1.8	0.3566
0.65	5.525	1.755	1.1744	2.9294	-1.95	0.3863
0.7	5.95	1.69	1.2647	2.9547	-2.1	0.4161
0.75	6.375	1.625	1.355	2.98	-2.25	0.4458
0.8	6.8	1.56	1.4454	3.0054	-2.4	0.4755
0.85	7.225	1.495	1.5357	3.0307	-2.55	0.5052
0.9	7.65	1.43	1.6261	3.0561	-2.7	0.5349
0.95	8.075	1.365	1.7164	3.0814	-2.85	0.5647
1.0	8.5	1.3	1.8067	3.1067	-3.0	0.5944

(columns 2-7 of 8; eta repeated on each block)

eta	Re_local
0.0	8246200.0
0.05	8040000.0
0.1	7833900.0
0.15	7627700.0
0.2	7421600.0
0.25	7215400.0
0.3	7009300.0
0.35	6803100.0
0.4	6597000.0
0.45	6390800.0
0.5	6184700.0
0.55	5978500.0
0.6	5772300.0
0.65	5566200.0
0.7	5360000.0
0.75	5153900.0
0.8	4947700.0
0.85	4741600.0
0.9	4535400.0
0.95	4329300.0
1.0	4123100.0

Table 8. Wing planform stations (wing_planform.csv). (columns 8-8 of 8; the table is split across 2 blocks so that no cell is compressed to illegibility, with eta repeated on each)

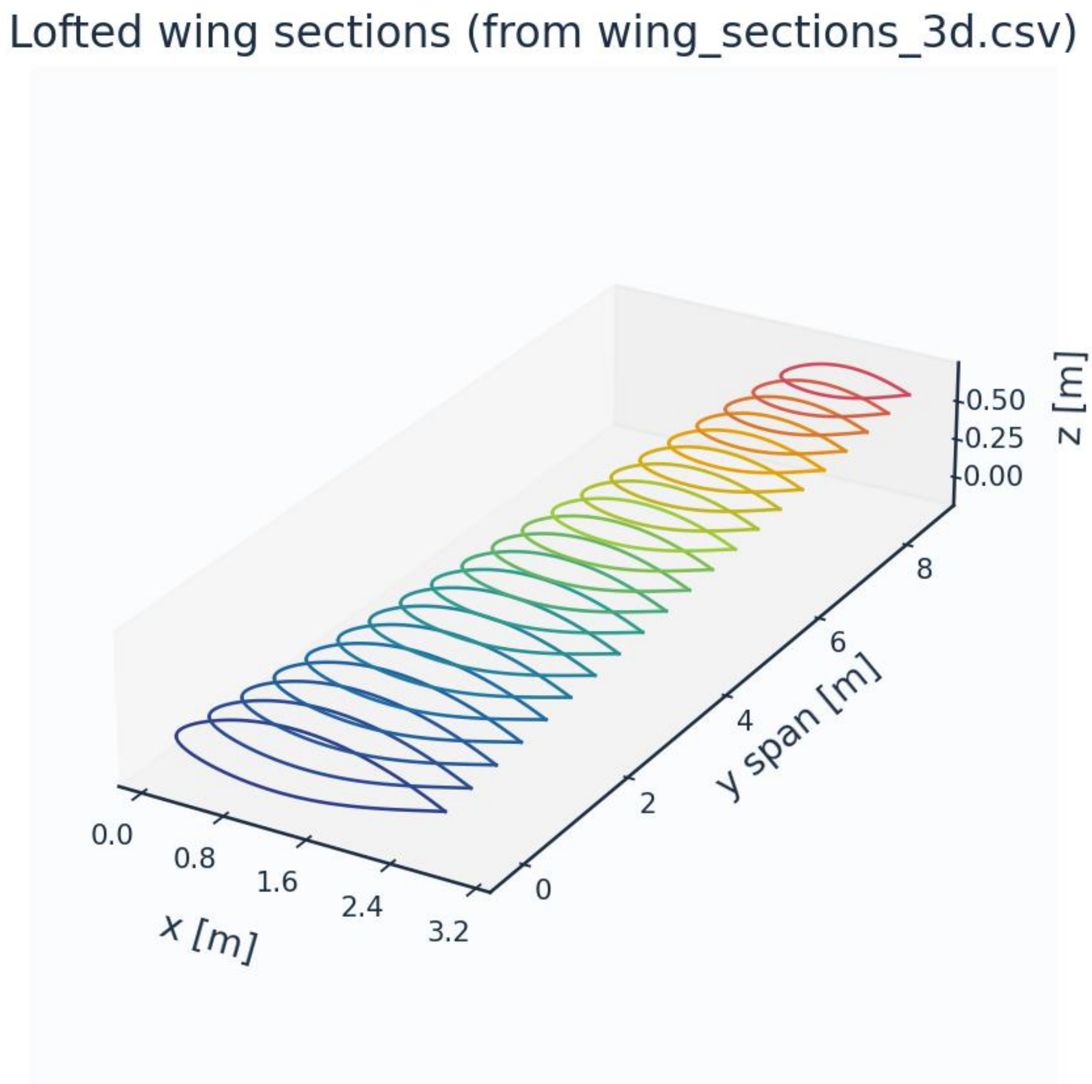


Fig. 11. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

THERE IS NO VOLUME MESH. This method is a surface panel discretisation coupled to an integral boundary layer, and the wall-normal stack below is a reconstruction grid used to recover profiles from the marched integral quantities - it is not a grid any equation is solved on. The y^+, growth ratio and aspect ratio in Table 9 are reported because they characterise that reconstruction, and they are the quantities a reader coming from a finite-volume solver will look for; they should not be read as evidence of a resolved near-wall grid, because there is none to resolve. This paragraph used to say so only in the last row of the table.

The surface is discretised with cosine-clustered streamwise nodes; the wall-normal reconstruction grid places its first point at $y^+ = 0.80$. A panel-count sweep bounds the discretisation sensitivity rather than demonstrating asymptotic convergence: from 120 to 700 surface panels the section drag spans 3.6 counts, 7.8 % of its mean, and above 180 panels it stays within about ±1.5 count without tightening further. The residual wander is not a truncation error that refinement removes — it is set by which panel the transition point lands on, so it scales with the panel spacing at transition and is of the same order as the ±0.025c bracket the aerofoil measurements themselves carry. The 260-panel grid is used for every case-study result; the tabulated validation sections are re-splined onto their own cosine-clustered grids of 400 and 440 panels.

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	1e-3 c
Max streamwise spacing	9.927	1e-3 c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wall-normal reconstruction stack (no volume mesh is generated)	-

Table 9. Metrics of the surface discretisation and of the wall-normal reconstruction stack. No volume mesh is generated.

n_surface_panels	Cl	Cd	x_tr_upper_c	dCd_pct
120	0.49385	0.0043266	0.539	
180	0.49531	0.004386	0.5434	1.373
260	0.49617	0.0045049	0.542	2.711
360	0.4967	0.0045745	0.539	1.544
480	0.49705	0.0046551	0.5357	1.761
600	0.49726	0.0046615	0.539	0.139
700	0.49737	0.0046823	0.5379	0.445

Table 10. Panel-count sensitivity study.

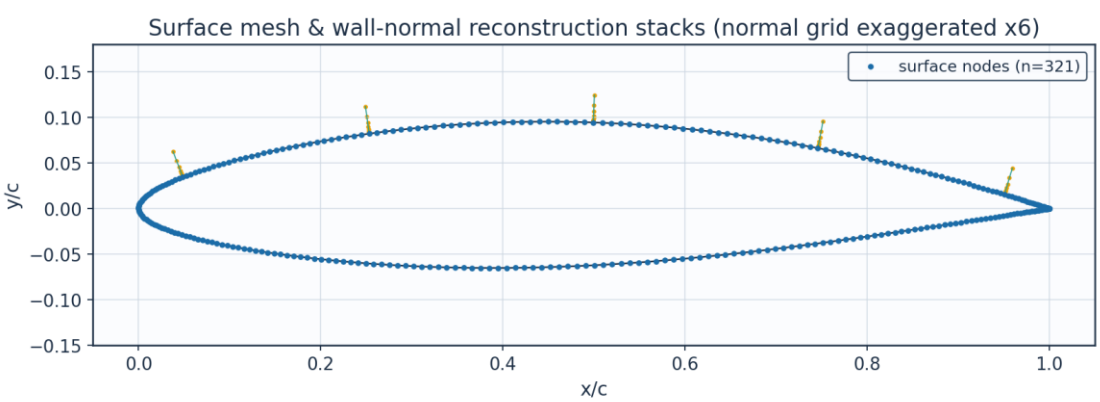


Fig. 12. Surface mesh and wall-normal stacks.

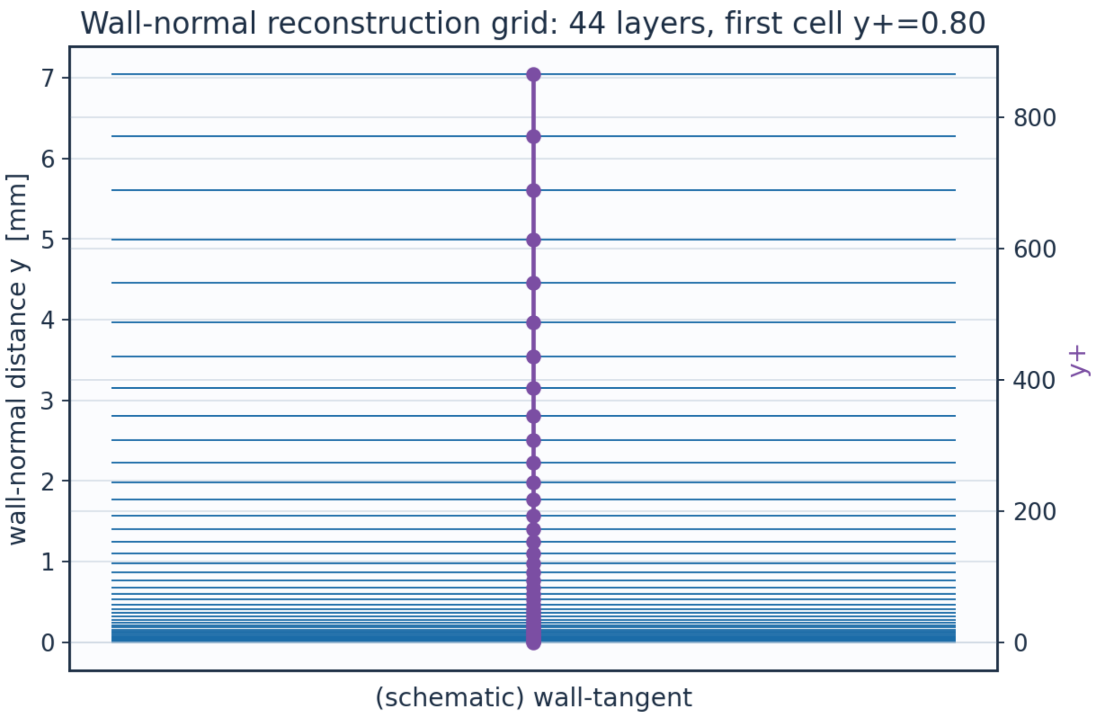


Fig. 13. Wall-normal reconstruction grid / y^+.

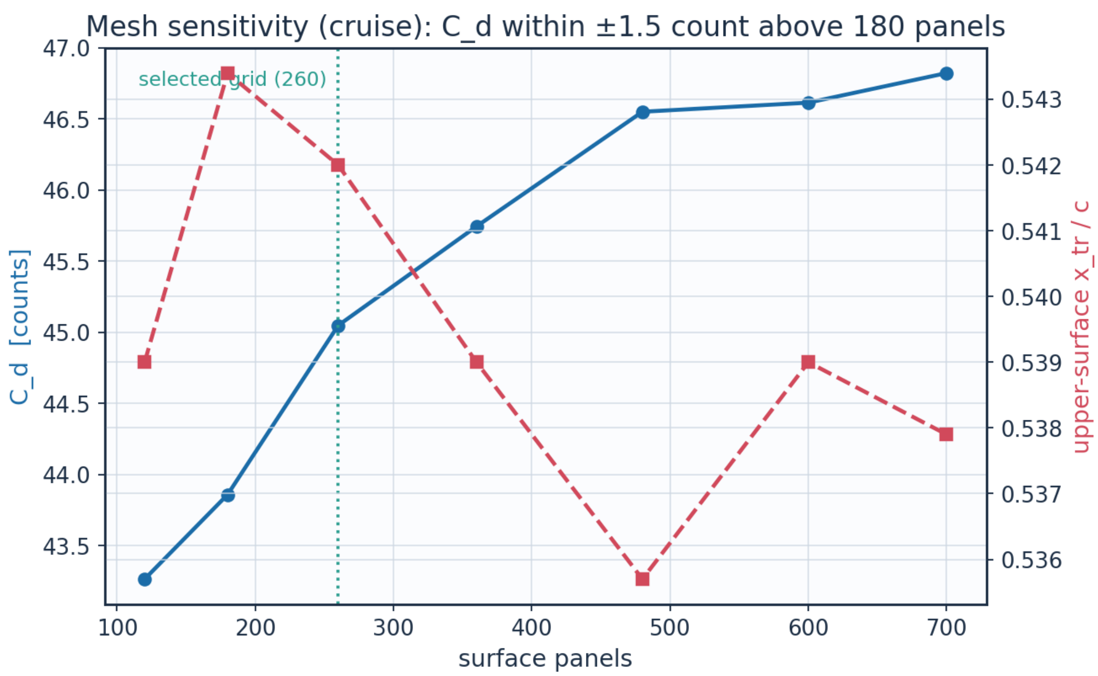


Fig. 14. Panel-count sensitivity of C_d and transition location.

layer	y_mm	y_plus	cell_dy_mm	growth_ratio
0	0.0	0.0	0.00651	1.0
2	0.0138	1.696	0.00773	1.12

4	0.0311	3.8235	0.0097	1.12
6	0.0529	6.4922	0.01217	1.12
8	0.0801	9.8398	0.01526	1.12
10	0.1143	14.039	0.01914	1.12
12	0.1572	19.3065	0.02401	1.12
14	0.211	25.9141	0.03012	1.12
16	0.2784	34.2026	0.03779	1.12
18	0.3631	44.5998	0.0474	1.12
20	0.4693	57.642	0.05946	1.12
22	0.6024	74.0021	0.07458	1.12
24	0.7695	94.5242	0.09356	1.12
26	0.9791	120.2671	0.11736	1.12
28	1.242	152.5591	0.14722	1.12
30	1.5717	193.0661	0.18467	1.12
32	1.9854	243.8782	0.23165	1.12
34	2.5043	307.6168	0.29058	1.12
36	3.1552	387.5705	0.3645	1.12
38	3.9717	487.8644	0.45723	1.12
40	4.9959	613.6731	0.57355	1.12
42	6.2806	771.4876	0.71946	1.12

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled to 22 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_ onset	mechanism
CRUISE	upper	0.554	0.542	3.554e+06	1380.4	TS-natural
CRUISE	lower	0.563	0.566	3.609e+06	1224.1	TS-natural
CLIMB	upper	0.046	0.025	4.958e+05	484.8	bypass
CLIMB	lower	0.165	0.166	1.762e+06	611.7	bypass

(columns 2-7 of 13; case repeated on each block)

case	laminar_run_ pct	Cf_te	H_te	H_te_at_clip	sep_margin_H	x_sep_turb_c
CRUISE	54.2	0.0002	2.801	True	-0.201	0.987
CRUISE	56.6	0.00024	2.802	True	-0.202	0.997
CLIMB	2.5	0.00014	2.8	True	-0.2	0.978
CLIMB	16.6	0.00019	2.8	True	-0.2	0.998

Table 12. Transition prediction summary (transition_summary.csv). (columns 8-13 of 13; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

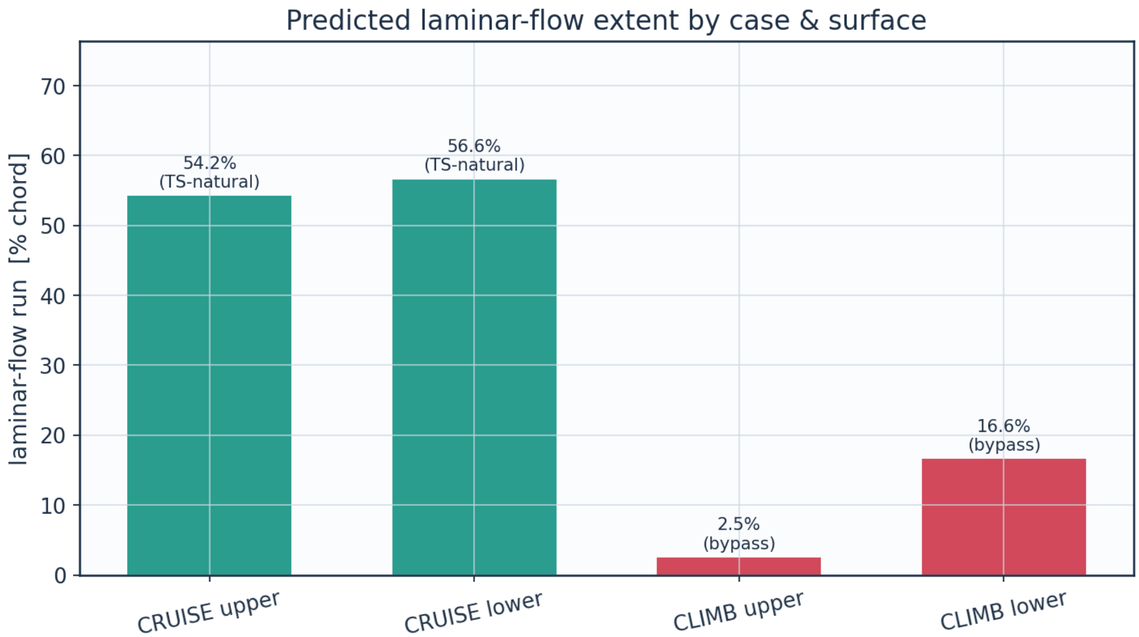


Fig. 15. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

The wing lift is obtained from Prandtl's lifting-line theory - Glauert's monoplane equation solved for the odd Fourier coefficients of the loading - using the section lift-curve slope and zero-lift incidence returned by the same panel method, the planform chord distribution, the wing's washout, and the section slope reduced by the cosine of the quarter-chord sweep. An earlier version of this table asserted C_L = 0.90 c_l; the planform actually returns 0.49, because a −3° washout is large relative to the 3.7° by which the root exceeds its zero-lift incidence, and because the downwash of an AR = 8.7 wing reduces the slope by a quarter. The span efficiency and the induced drag come out of the same solve. The implementation is checked against the one case with a closed-form answer - an elliptic planform, for which it returns e = 1 and the exact lift-curve slope to machine precision - every time the solution is regenerated.

The incidence in Table 4 is the SECTION design incidence, not the trim point of the aircraft the planform belongs to: at 1.5° the wing carries 23.6 kN against an all-up weight of 83.4 kN, and level flight at MTOW would need 7.6°. Table 13 reports both, so the two are not confused. Every transition result in this study is quoted at the design incidence and is unaffected by that distinction.

quantity	value	unit
Section lift coefficient Cl	0.4962	-
Section profile drag Cd	0.0045	-
Section Cd (counts)	45.0	counts
Section L/D	110.1	-
Section lift-curve slope a0 (panel)	8.111	1/rad
Zero-lift incidence a_L0 (panel)	-2.185	deg
Quarter-chord sweep	9.89	deg
Wing C_L (lifting line, taper + washout + sweep)	0.2548	-

Wing C_L / section c_l	0.513	-
Span efficiency e (lifting line)	0.8718	-
Wing induced drag C_Di (lifting line)	0.00272	-
Wing lift (lifting line)	23603.0	N
Wing C_L for level flight at MTOW	0.8998	-
Incidence for that C_L (lifting line)	7.55	deg
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6.414e+06	-
Mach number	0.42	-

Table 13. Integrated forces, with the wing quantities from the lifting-line solve.

configuration	Cd_counts	mean_laminar_pct	viscous_drag_reduction_pct
NLF (UTSS predicted transition)	45.0	55.4	50.3
Fully turbulent (LE trip)	90.6	0.0	0.0

Table 14. NLF vs fully-turbulent drag.

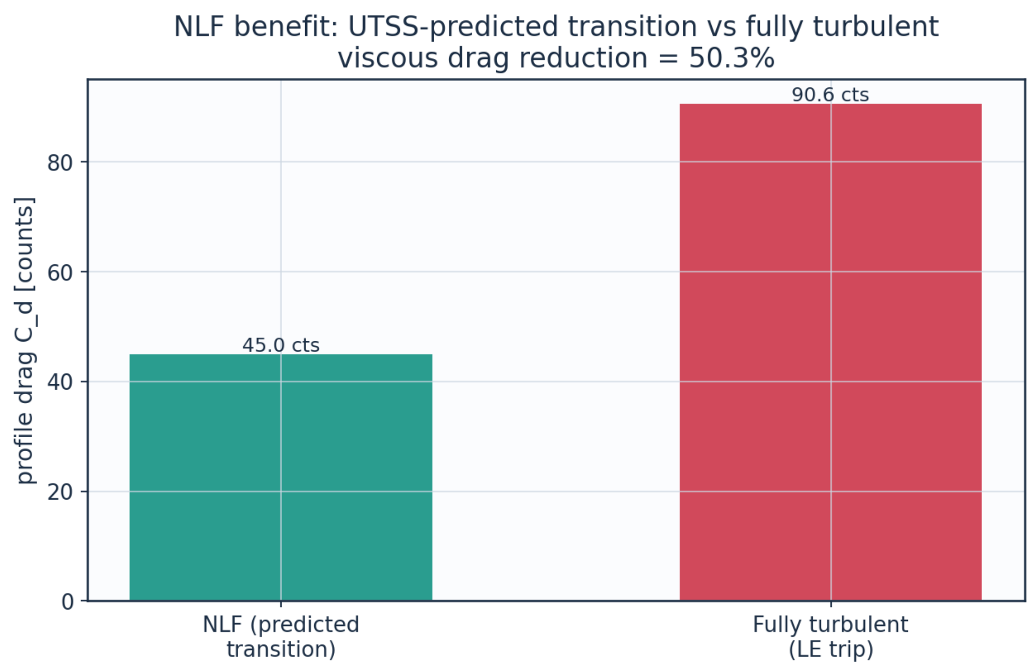


Fig. 16. Drag benefit of predicted laminar flow vs fully-turbulent.

9.2a The transition-length closure and what the result owes to it

The extent of the transitional region is Dhawan and Narasimha's published correlation, $Re_\lambda = 9\, Re_{x,t}^{0.75}$, with λ the distance over which the intermittency rises from 0.25 to 0.75. This work previously reported it as validated on the flat plates of Section 11.1, which span $Re_{x,t} = 6\times10^4$ to 1.4×10^6, and EXTRAPOLATED on the wing, which transitions at 3.6×10^6. That extrapolation was of the variable, not of the physics.

Their correlation is Narasimha's spot model with a constant dimensionless spot formation rate. Matching $\gamma = 1 - \exp(-0.412\,\xi^2)$ against $\gamma = 1 - \exp(-n\,\sigma\,(x-x_t)^2/U)$, with $\hat{N} = n\,\sigma\,\theta_t^3/\nu$, gives $Re_\lambda = \sqrt{0.412/\hat{N}}\; Re_{\theta,t}^{1.5}$; and on a Blasius plate $Re_\theta = 0.664\sqrt{Re_x}$, so $Re_\theta^{1.5} = 0.5411$

Re_x^0.75 and the two are THE SAME LAW with C = 9/0.5411 = 16.63. Checked over Re_x from 6×10^4 to 10^7 they agree to every figure printed. What is assumed constant is therefore the spot formation rate, and Re_θ is the variable in which that assumption is stated exactly. Nothing is fitted and nothing is extrapolated: the constant is still their published 9.0.

Measured, the two forms agree to 0.03 per cent on the four zero-pressure-gradient plates — as they must, since Re_θ = 0.664 √Re_x there — and differ by 2.85 per cent on T3C4, the one plate that carries a pressure gradient, where the Re_θ form is the correct one. On the cruise wing the transitional length moves by 0.4 per cent and the section drag by 0.02 counts. The 0.02 counts is not the point. The point is that the wing no longer sits outside the range of a correlation; it sits inside the range of an assumption that was always that correlation's actual content. cal["len_re_x"] recovers the published form so the equivalence can be checked rather than taken on trust.

What the case-study drag owes to the closure is measured all the same, by sweeping the constant over a factor of four:

C_len	multiple_of_published	Cd_counts	x_tr_c_upper	transitional_extent_pct_chord	completes_before_TE
2.25	0.25	45.09	0.542	8.4	True
4.5	0.5	45.09	0.542	18.5	True
9.0	1.0	45.05	0.542	36.2	True
18.0	2.0	42.13	0.542		False
36.0	4.0	36.76	0.542		False

Table 15. Sensitivity of the case-study result to the transition-length constant (transition_length_sensitivity.csv).

9.3 Surface distributions — cruise

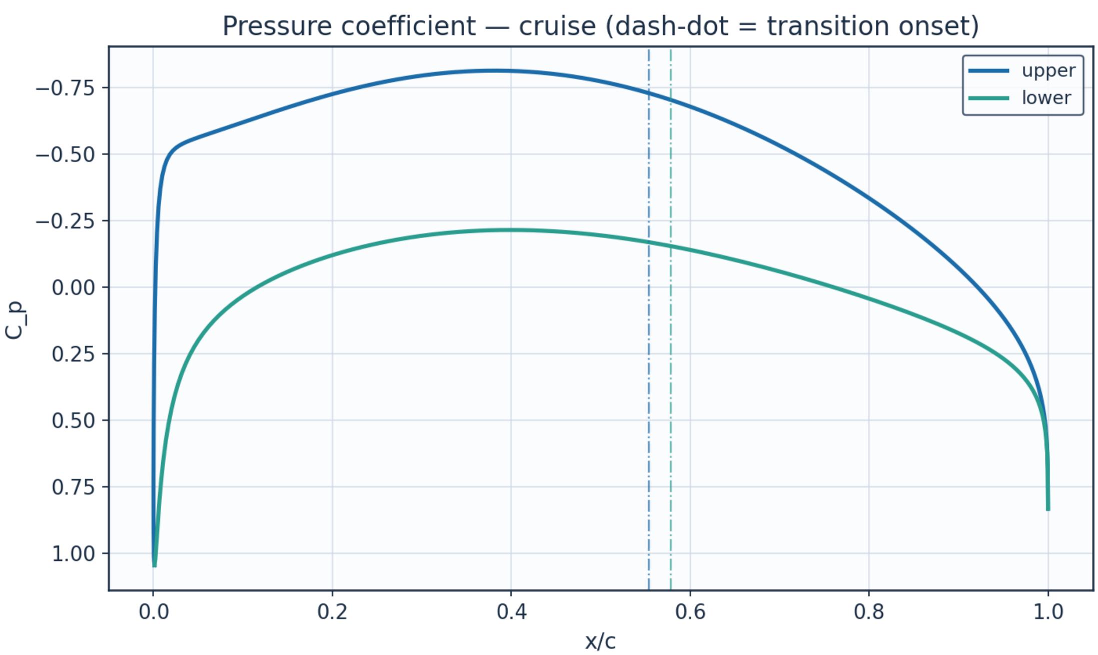


Fig. 17. Pressure coefficient C_p — cruise.

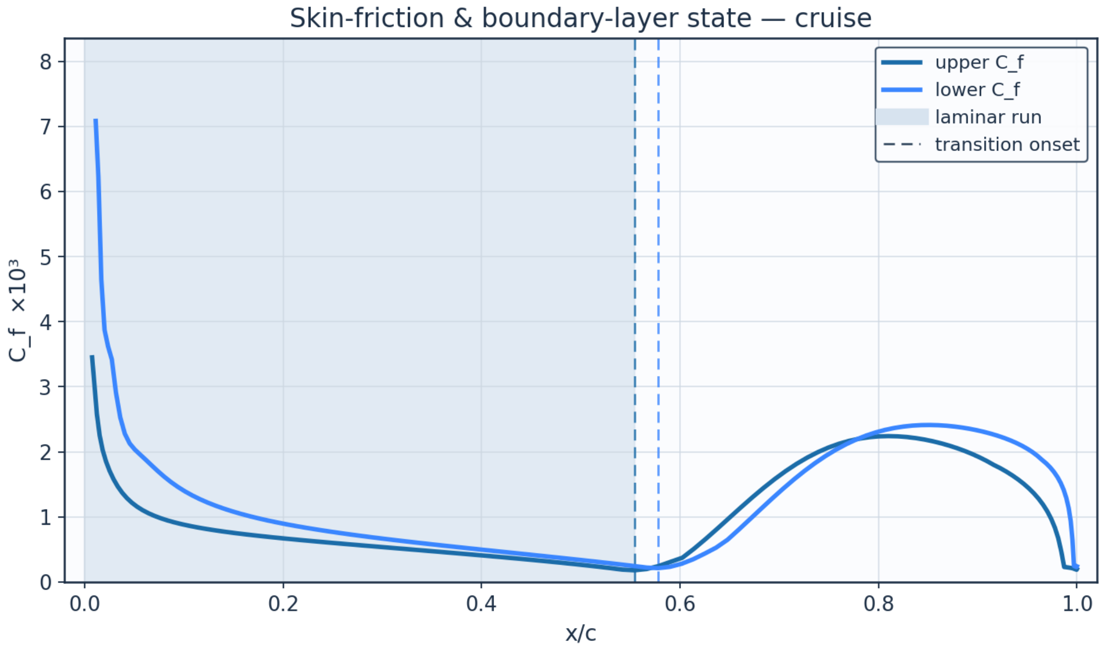


Fig. 18. Skin-friction C_f and laminar run — cruise.

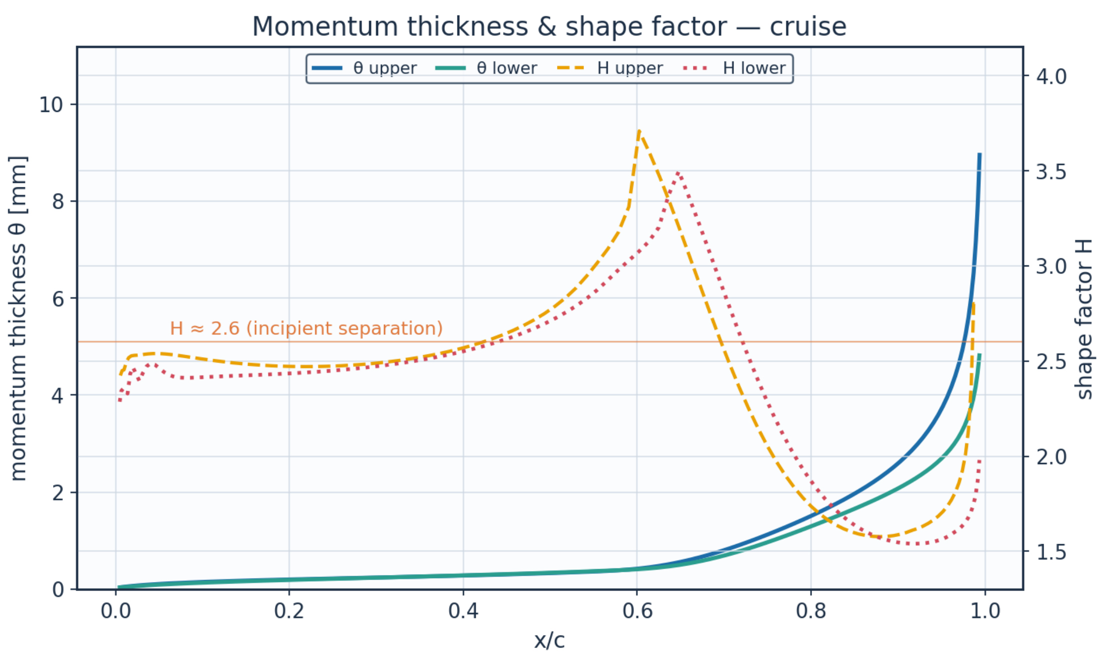


Fig. 19. Momentum thickness ϑ and shape factor H — cruise.

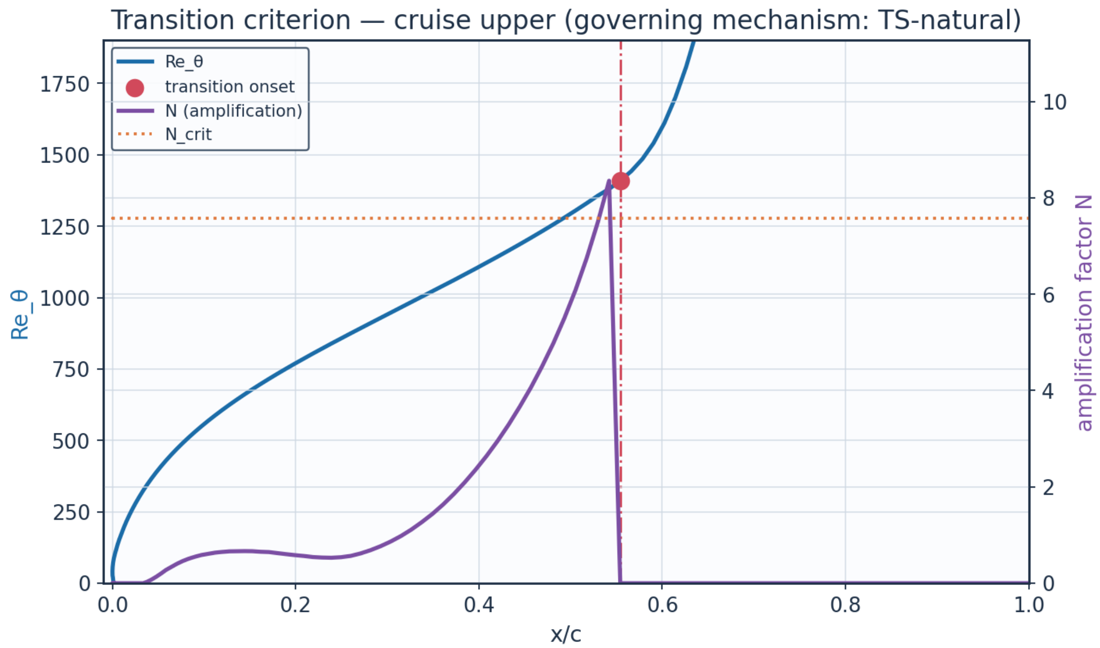


Fig. 20. The governing transition criterion — cruise. Two of the four branches are Reynolds-number thresholds and two are amplification integrals, so both pairs are drawn; at cruise it is N reaching N_crit that fires, and no Re_ϑt threshold is active at any station.

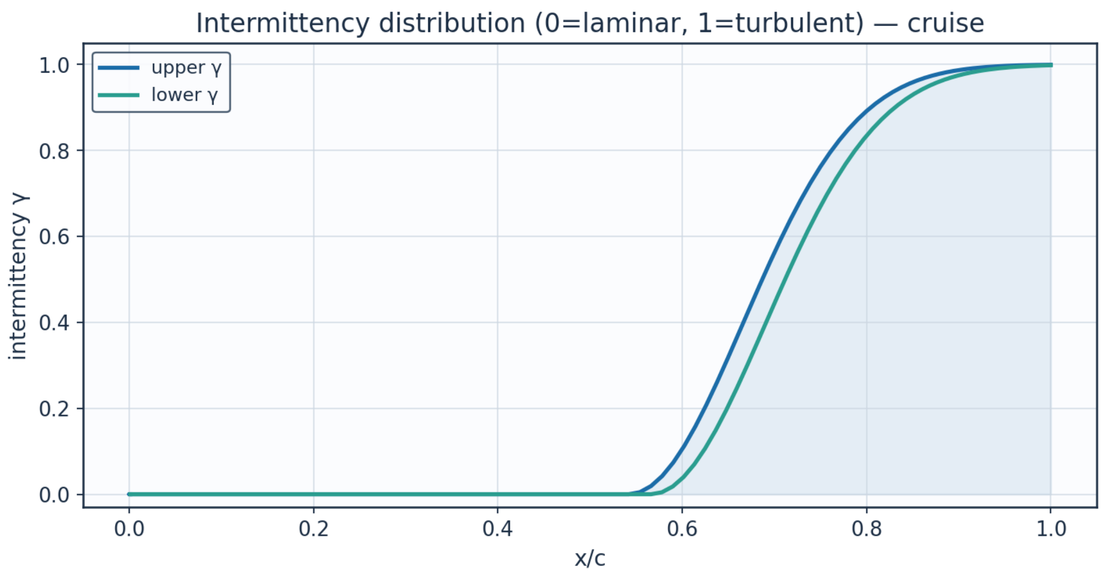


Fig. 21. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm
0.001152	0.0	0.0	1.04617	0.0001	0.0	1e-05
0.000243	0.01379	42779.0	0.51697	85.1945	0.68744	0.02686
0.003998	0.029694	92120.0	-0.20578	133.2104	1.07489	0.03613
0.012383	0.051716	160438.0	-0.45363	146.6018	1.18295	0.05492
0.025317	0.08137	252436.0	-0.52328	150.2001	1.21198	0.07747
0.042683	0.119051	369334.0	-0.55314	151.7229	1.22427	0.10042

0.064323	0.164707	510973.0	-0.57855	153.0105	1.23466	0.12236
0.090042	0.218063	676500.0	-0.60768	154.4766	1.24649	0.14279
0.119612	0.278705	864629.0	-0.64128	156.1558	1.26004	0.16174
0.152769	0.346116	1073758.0	-0.6778	157.9661	1.27465	0.17949
0.189218	0.419703	1302050.0	-0.71481	159.7856	1.28933	0.19642
0.228631	0.498815	1547479.0	-0.74951	161.4789	1.30299	0.21299
0.27065	0.582752	1807876.0	-0.77904	162.9098	1.31454	0.22971
0.31489	0.670778	2080961.0	-0.80063	163.9512	1.32294	0.24714
0.360938	0.762129	2364360.0	-0.81192	164.4934	1.32732	0.26585
0.408355	0.856016	2655625.0	-0.81105	164.4517	1.32698	0.28651
0.456686	0.951626	2952238.0	-0.7969	163.7712	1.32149	0.30983
0.505459	1.048128	3251618.0	-0.76911	162.4298	1.31066	0.33671
0.554191	1.144672	3551128.0	-0.72812	160.4365	1.29458	0.36851
0.6024	1.240392	3848080.0	-0.675	157.8277	1.27353	0.42729
0.649607	1.334412	4139757.0	-0.61134	154.6601	1.24797	0.56189
0.695348	1.425852	4423433.0	-0.53899	151.0029	1.21846	0.7787
0.739176	1.513841	4696404.0	-0.45992	146.9295	1.18559	1.05223
0.780671	1.597528	4956026.0	-0.37596	142.51	1.14993	1.3554
0.819445	1.676092	5199755.0	-0.28872	137.8042	1.11196	1.67714
0.855143	1.748757	5425183.0	-0.19946	132.8555	1.07203	2.02313
0.887448	1.814803	5630080.0	-0.10897	127.6845	1.0303	2.41041
0.916081	1.873579	5812420.0	-0.01756	122.2815	0.9867	2.8652
0.940801	1.924505	5970409.0	0.07514	116.593	0.9408	3.42777
0.961407	1.967085	6102504.0	0.17034	110.4954	0.8916	4.1686

(columns 2-7 of 15; x_c repeated on each block)

x_c	H_shape	Cf	Re_theta	Re_theta_ trans	n_factor	n_crit
0.001152	2.61	16804859.1563012	0.0		0.0	7.584
0.000243	2.2504	0.0120106	57.47		0.0	7.584
0.003998	2.346	0.0051896	117.58		0.0	7.584
0.012383	2.493	0.0025849	194.69		0.0	7.584
0.025317	2.5306	0.0017063	280.56		0.0	7.584
0.042683	2.539	0.0012901	366.91		0.0761	7.584
0.064323	2.5337	0.0010585	450.4		0.3393	7.584
0.090042	2.5171	0.0009195	529.97		0.5497	7.584
0.119612	2.4986	0.0008244	605.98		0.6494	7.584
0.152769	2.4829	0.0007509	679.21		0.6645	7.584
0.189218	2.4731	0.0006884	750.63		0.6039	7.584
0.228631	2.4709	0.000631	821.34		0.5329	7.584
0.27065	2.478	0.0005753	892.53		0.6176	7.584
0.31489	2.4961	0.0005191	965.45		0.9798	7.584
0.360938	2.5284	0.000461	1041.5		1.6351	7.584
0.408355	2.5795	0.0003994	1122.17		2.6438	7.584
0.456686	2.6594	0.0003324	1209.25		4.0628	7.584
0.505459	2.7874	0.0002571	1304.96		6.0907	7.584
0.554191	3.0161	0.0001815	1409.18		0.0	7.584
0.6024	3.7092	0.0003742	1611.22		0.0	7.584
0.649607	3.1848	0.0009861	2082.12		0.0	7.584
0.695348	2.613	0.0015816	2826.32		0.0	7.584

0.739176	2.1455	0.0019981	3728.9		0.0	7.584
0.780671	1.8353	0.0022031	4675.7		0.0	7.584
0.819445	1.6642	0.0022389	5615.4		0.0	7.584
0.855143	1.5902	0.0021619	6555.21		0.0	7.584
0.887448	1.578	0.002011	7534.4		0.0	7.584
0.916081	1.6095	0.0018007	8609.32		0.0	7.584
0.940801	1.6642	0.0015755	9857.67		0.0	7.584
0.961407	1.7689	0.0012795	11404.73		0.0	7.584

(columns 8-13 of 15; x_c repeated on each block)

x_c	intermittency_gamma	state
0.001152	0.0	laminar
0.000243	0.0	laminar
0.003998	0.0	laminar
0.012383	0.0	laminar
0.025317	0.0	laminar
0.042683	0.0	laminar
0.064323	0.0	laminar
0.090042	0.0	laminar
0.119612	0.0	laminar
0.152769	0.0	laminar
0.189218	0.0	laminar
0.228631	0.0	laminar
0.27065	0.0	laminar
0.31489	0.0	laminar
0.360938	0.0	laminar
0.408355	0.0	laminar
0.456686	0.0	laminar
0.505459	0.0	laminar
0.554191	0.00482	transitional
0.6024	0.11268	transitional
0.649607	0.31667	transitional
0.695348	0.53979	transitional
0.739176	0.72416	transitional
0.780671	0.84959	transitional
0.819445	0.92346	transitional
0.855143	0.96261	transitional
0.887448	0.98193	transitional
0.916081	0.9911	turbulent
0.940801	0.9954	turbulent
0.961407	0.99744	turbulent

Table 16. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled). (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with x_c repeated on each)

(table sampled to 30 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm
0.001152	0.0	0.0	1.04617	0.0001	0.0	1e-05
0.006721	0.017175	53283.0	0.80073	57.4943	0.46393	0.04058
0.016897	0.041011	127228.0	0.51407	85.4337	0.68937	0.05476
0.031587	0.07262	225291.0	0.32494	99.9191	0.80626	0.07452
0.05065	0.112235	348189.0	0.19779	108.6835	0.87698	0.0956
0.073903	0.159719	495497.0	0.10481	114.7223	0.92571	0.11572
0.101119	0.214733	666169.0	0.03157	119.2949	0.9626	0.13601

0.132034	0.276811	858752.0	-0.02902	122.97	0.99226	0.15583
0.166346	0.345385	1071492.0	-0.08027	126.0087	1.01678	0.17517
0.203719	0.419819	1302410.0	-0.12345	128.5229	1.03707	0.19422
0.243794	0.499421	1549360.0	-0.15873	130.5483	1.05341	0.21329
0.286186	0.583462	1810078.0	-0.18577	132.0838	1.0658	0.23276
0.330494	0.671185	2082224.0	-0.20407	133.1145	1.07412	0.25303
0.376307	0.761822	2363408.0	-0.2132	133.6266	1.07825	0.27457
0.423206	0.854593	2651213.0	-0.21303	133.6172	1.07817	0.29785
0.470769	0.948713	2943202.0	-0.20381	133.1	1.074	0.32338
0.518574	1.043391	3236922.0	-0.18618	132.1069	1.06598	0.3517
0.566202	1.13783	3529900.0	-0.16115	130.6864	1.05452	0.38345
0.613236	1.231224	3819638.0	-0.12998	128.8999	1.04011	0.42926
0.659265	1.322763	4103619.0	-0.09405	126.8151	1.02328	0.53301
0.703882	1.411629	4379309.0	-0.05469	124.4997	1.0046	0.71314
0.746688	1.497007	4644178.0	-0.01313	122.0148	0.98455	0.94915
0.787291	1.57809	4895723.0	0.02971	119.409	0.96352	1.21014
0.825311	1.65409	5131498.0	0.07322	116.7134	0.94177	1.47516
0.860385	1.724249	5349152.0	0.11716	113.9362	0.91936	1.73825
0.892165	1.787851	5546467.0	0.16175	111.0573	0.89613	2.00457
0.920332	1.844238	5721396.0	0.20775	108.0196	0.87162	2.28651
0.944596	1.892815	5872098.0	0.25655	104.7133	0.84494	2.60448
0.964701	1.933066	5996968.0	0.31058	100.9425	0.81452	2.99535
0.980432	1.964558	6094667.0	0.37411	96.3421	0.77739	3.53851

(columns 2-7 of 15; x_c repeated on each block)

x_c	**H_shape**	**Cf**	**Re_theta**	**Re_theta_ trans**	**n_factor**	**n_crit**
0.001152	2.61	16804858.4967999	0.0		0.0	7.584
0.006721	2.343	0.0103475	59.21		0.0	7.584
0.016897	2.4316	0.0046449	117.49		0.0	7.584
0.031587	2.4376	0.0029162	185.68		0.0	7.584
0.05065	2.4611	0.0020358	257.87		0.0	7.584
0.073903	2.4128	0.0017038	328.32		0.0	7.584
0.101119	2.4149	0.0013943	400.12		0.0	7.584
0.132034	2.4213	0.0011735	471.44		0.0	7.584
0.166346	2.4275	0.0010124	541.94		0.0	7.584
0.203719	2.4356	0.0008874	611.82		0.0	7.584
0.243794	2.4473	0.0007844	681.53		0.0	7.584
0.286186	2.4649	0.0006949	751.65		0.0272	7.584
0.330494	2.4904	0.0006136	822.89		0.317	7.584
0.376307	2.5269	0.0005369	896.04		0.9481	7.584
0.423206	2.5781	0.000462	971.96		1.9344	7.584
0.470769	2.6512	0.0003866	1051.58		3.2983	7.584
0.518574	2.7571	0.0003084	1135.95		5.1195	7.584
0.566202	2.9248	0.0002234	1224.09		7.7054	7.584
0.613236	3.1268	0.0003506	1353.37		0.0	7.584
0.659265	3.3665	0.0008073	1655.75		0.0	7.584
0.703882	2.8127	0.0014346	2178.39		0.0	7.584
0.746688	2.317	0.0019373	2846.28		0.0	7.584

0.787291	1.9555	0.0022491	3557.65		0.0	7.584
0.825311	1.7304	0.0023897	4246.35		0.0	7.584
0.860385	1.6085	0.0024098	4893.27		0.0	7.584
0.892165	1.5534	0.0023539	5510.23		0.0	7.584
0.920332	1.5395	0.0022469	6124.53		0.0	7.584
0.944596	1.5543	0.0020942	6775.81		0.0	7.584
0.964701	1.5993	0.0018786	7528.06		0.0	7.584
0.980432	1.6732	0.0016112	8508.97		0.0	7.584

(columns 8-13 of 15; x_c repeated on each block)

x_c	intermittency_gamma	state
0.001152	0.0	laminar
0.006721	0.0	laminar
0.016897	0.0	laminar
0.031587	0.0	laminar
0.05065	0.0	laminar
0.073903	0.0	laminar
0.101119	0.0	laminar
0.132034	0.0	laminar
0.166346	0.0	laminar
0.203719	0.0	laminar
0.243794	0.0	laminar
0.286186	0.0	laminar
0.330494	0.0	laminar
0.376307	0.0	laminar
0.423206	0.0	laminar
0.470769	0.0	laminar
0.518574	0.0	laminar
0.566202	0.0	transitional
0.613236	0.06962	transitional
0.659265	0.24645	transitional
0.703882	0.46218	transitional
0.746688	0.65609	transitional
0.787291	0.79884	transitional
0.825311	0.88976	transitional
0.860385	0.94188	transitional
0.892165	0.96968	transitional
0.920332	0.9839	transitional
0.944596	0.99105	turbulent
0.964701	0.99466	turbulent
0.980432	0.9965	turbulent

Table 17. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled). (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with x_c repeated on each)

(table sampled to 30 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated Tu)

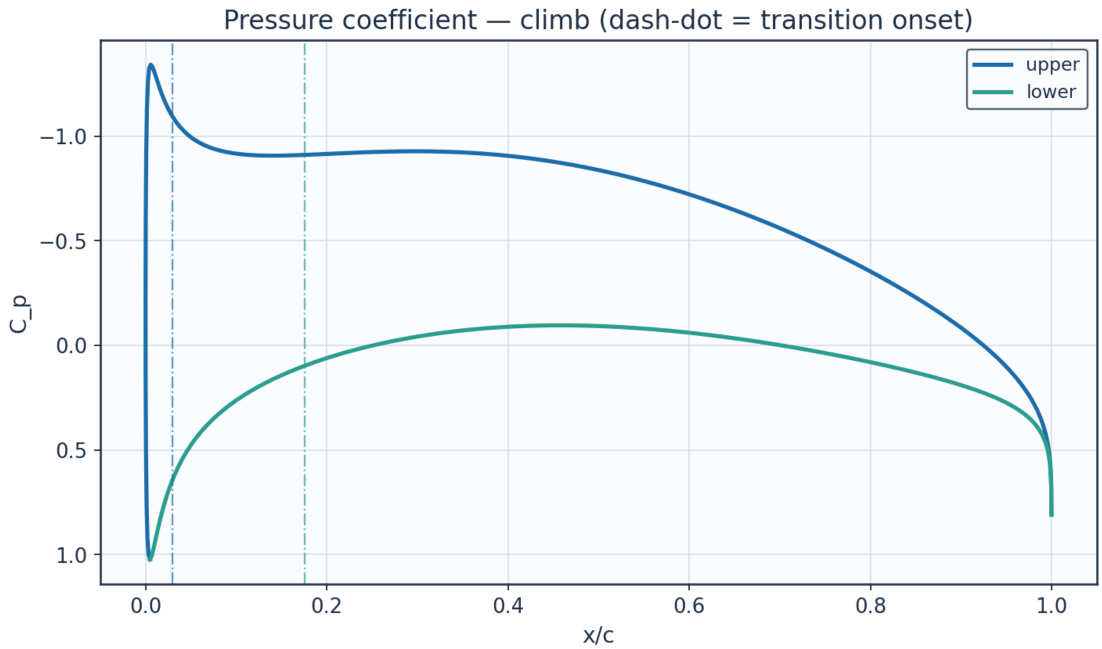


Fig. 22. Pressure coefficient C_p — climb.

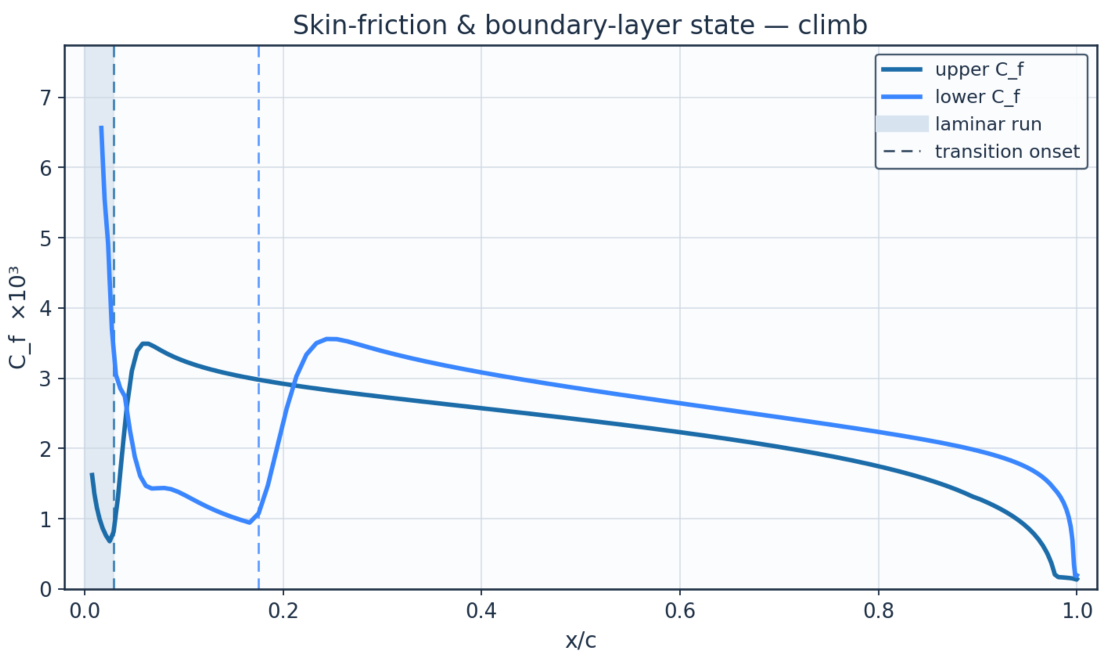


Fig. 23. Skin-friction C_f and laminar run — climb.

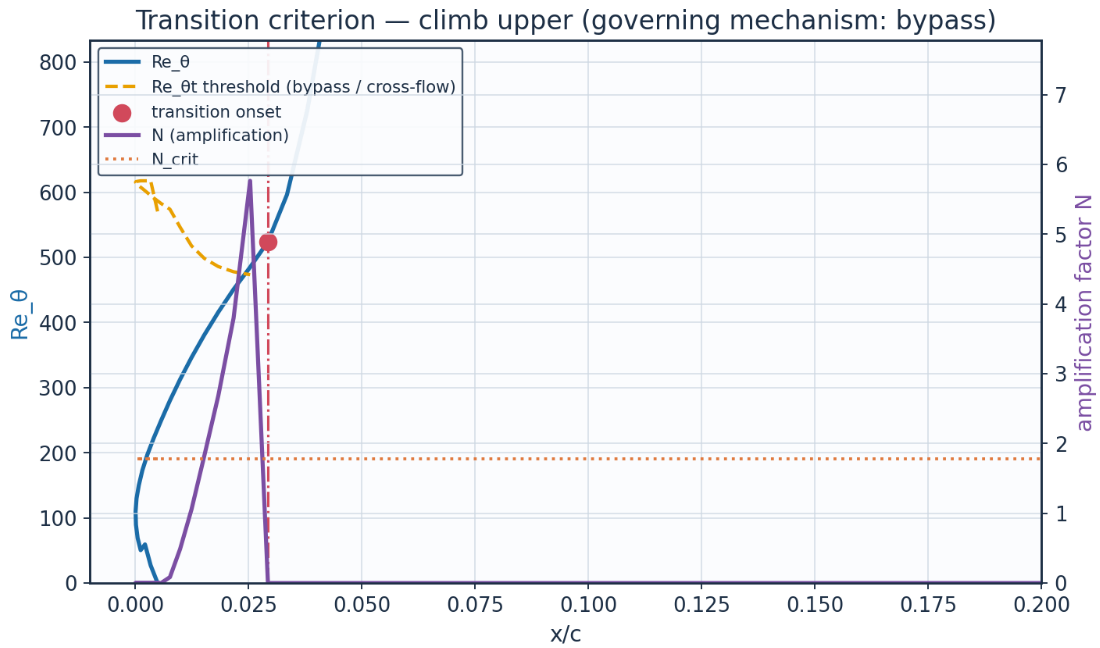


Fig. 24. The governing transition criterion — climb: here Re_ϑ crosses the falling Abu-Ghannam & Shaw bypass threshold while N is still far below N_crit.

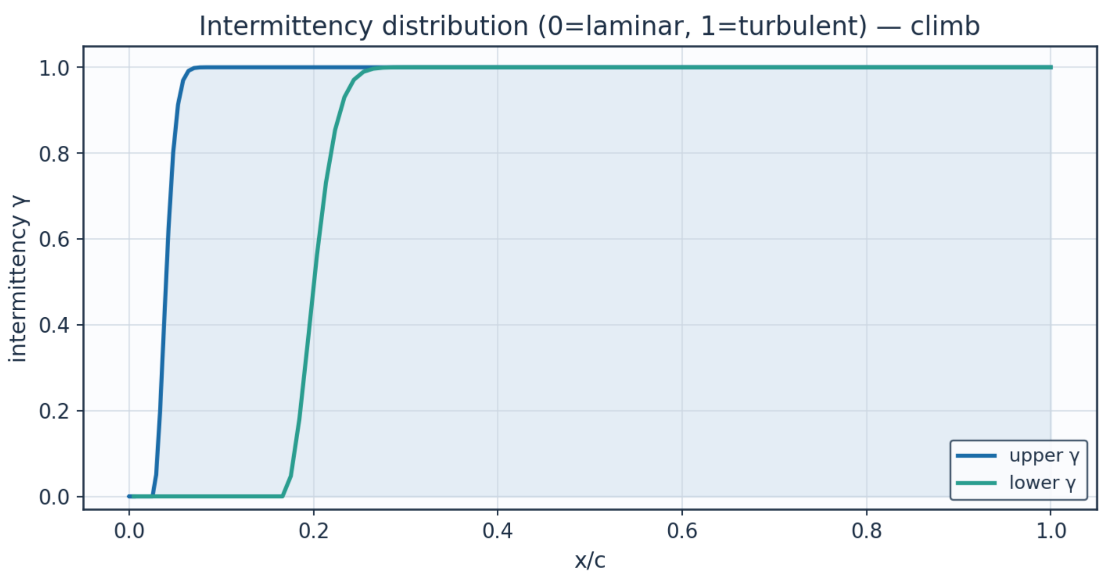


Fig. 25. Intermittency γ — climb.

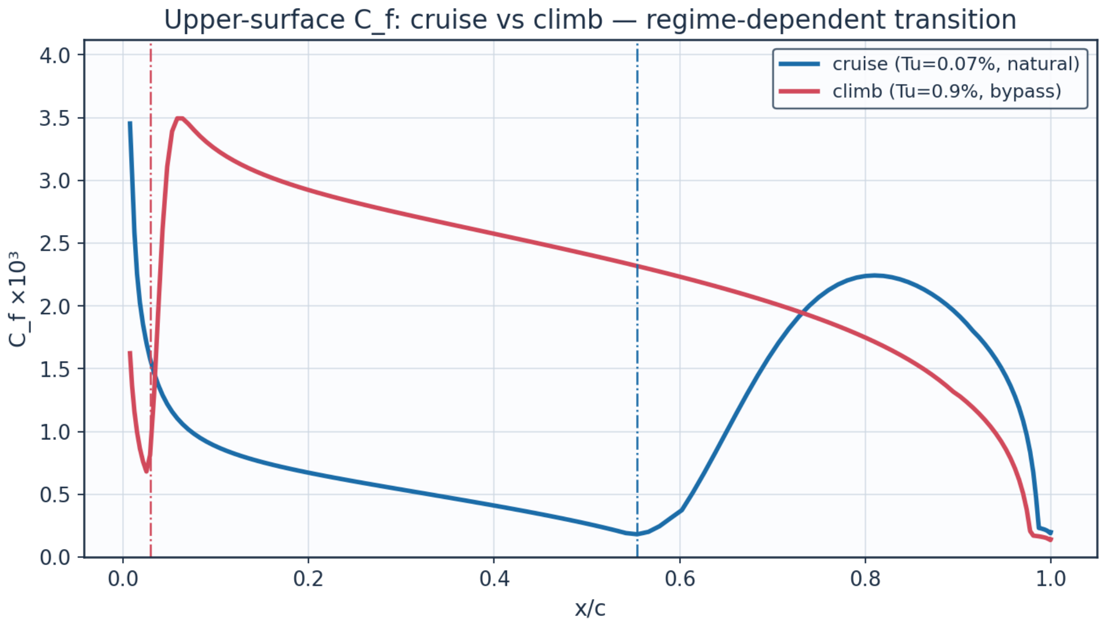


Fig. 26. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

The polar carries θ_TE/c, the trailing-edge momentum thickness as a fraction of chord, because that is the assumption Squire-Young rests on and it is the quantity that bounds where the drag can be believed. Over the sweep it stays below 0.009, so the thin-layer assumption holds throughout. Outside the incidence envelope it does not merely degrade: at 16° and a chord Reynolds number of $2{\times}10^5$ the march returns a momentum thickness of 1.34 chords and the formula duly returns C_d = 1.22, which is bluff-body drag from an aerofoil method. A layer thicker than the body is long is arithmetic that has stopped meaning anything rather than a marginal case, so the solver now returns no drag there instead of a number. Within the envelope and Reynolds range of this study the test never fires.

alpha_deg	Cl	Cd	L_over_D	theta_te_c	xtr_upper_c	xtr_lower_c
-3.0	-0.1149	0.00601	-19.1	0.00474	0.661	0.094
-2.0	0.0214	0.00479	4.5	0.00308	0.638	0.353
-1.0	0.1571	0.00443	35.4	0.00244	0.614	0.459
0.0	0.2926	0.00438	66.8	0.00237	0.59	0.507
1.0	0.4282	0.00445	96.3	0.00269	0.554	0.554
2.0	0.5643	0.00461	122.5	0.00306	0.518	0.59
3.0	0.7013	0.00753	93.1	0.00635	0.12	0.613
4.0	0.8396	0.00877	95.8	0.00778	0.033	0.648
5.0	0.9796	0.00974	100.5	0.00895	0.018	0.682
6.0	1.1219	0.01087	103.2	0.01024	0.018	0.704
7.0	1.2674	0.01259	100.7	0.01219	0.015	0.736
8.0	1.417	0.01625	87.2	0.01612	0.01	0.757

Table 18. Aerodynamic polar (aero_polar.csv).

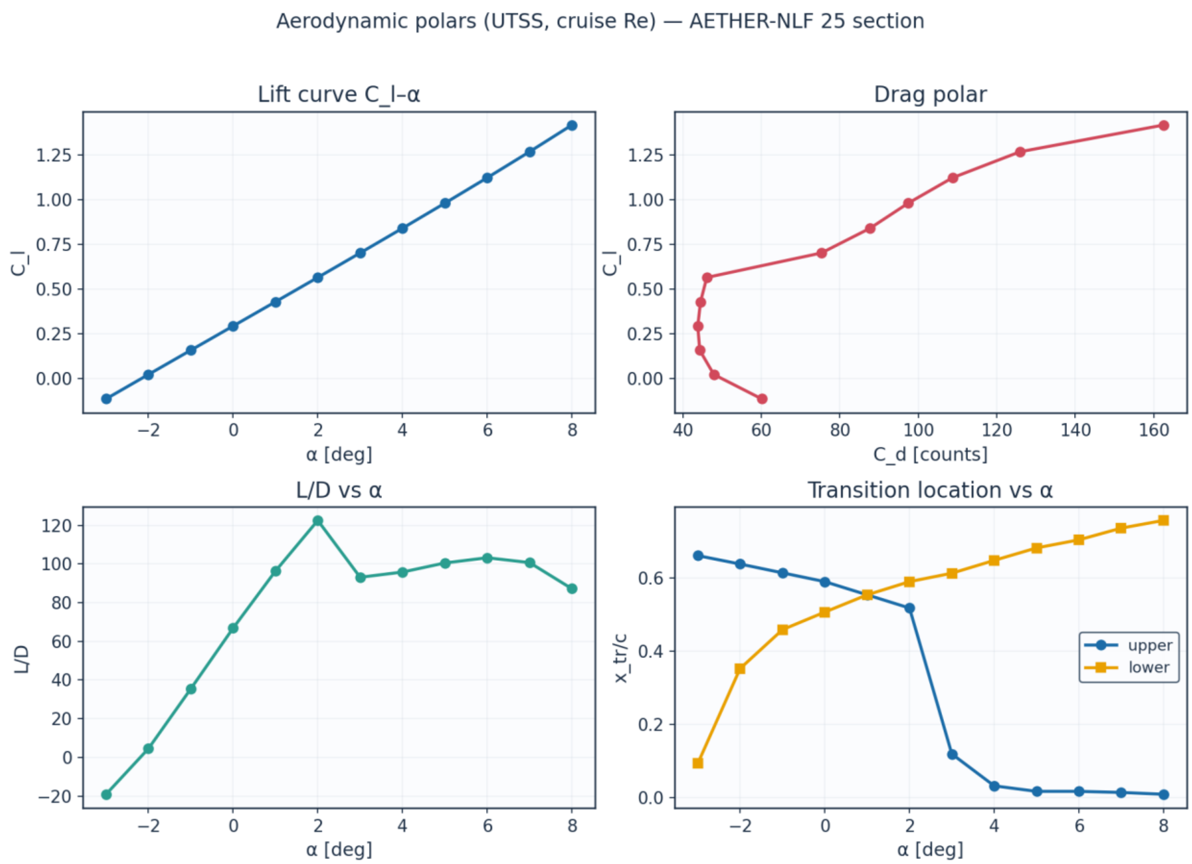


Fig. 27. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

eta	y_m	chord_m	Re_local	alpha_geom_deg	alpha_induced_deg	alpha_eff_deg
0.0	0.0	2.6	8246200.0	1.5	1.38	0.12
0.089	0.757	2.484	7878900.0	1.23	1.09	0.14
0.178	1.515	2.368	7511500.0	0.97	0.87	0.1
0.267	2.272	2.253	7144200.0	0.7	0.68	0.01
0.356	3.029	2.137	6776900.0	0.43	0.53	-0.1
0.445	3.786	2.021	6409500.0	0.16	0.4	-0.24
0.535	4.544	1.905	6042200.0	-0.1	0.29	-0.39
0.624	5.301	1.789	5674900.0	-0.37	0.19	-0.56
0.713	6.058	1.673	5307600.0	-0.64	0.12	-0.75
0.802	6.815	1.558	4940200.0	-0.91	0.07	-0.97
0.891	7.573	1.442	4572900.0	-1.17	0.07	-1.24
0.98	8.33	1.326	4205600.0	-1.44	0.28	-1.72

(columns 2-7 of 12; eta repeated on each block)

eta	c_l_section	xtr_upper_c	xtr_lower_c	Cd_section	laminar_fraction
0.0	0.3083	0.566	0.495	0.00431	0.53
0.089	0.3118	0.566	0.507	0.00431	0.536
0.178	0.3058	0.566	0.507	0.00436	0.536
0.267	0.2945	0.578	0.507	0.00434	0.543

0.356	0.2791	0.578	0.507	0.00439	0.543
0.445	0.2607	0.59	0.507	0.00438	0.549
0.535	0.2397	0.602	0.495	0.00442	0.549
0.624	0.2163	0.602	0.495	0.00448	0.549
0.713	0.1903	0.614	0.495	0.00449	0.554
0.802	0.1607	0.673	0.483	0.00433	0.578
0.891	0.1244	0.684	0.471	0.00443	0.577
0.98	0.059	0.695	0.447	0.0046	0.571

Table 19. Span-wise distribution (spanwise_distribution.csv). (columns 8-12 of 12; the table is split across 2 blocks so that no cell is compressed to illegibility, with eta repeated on each)

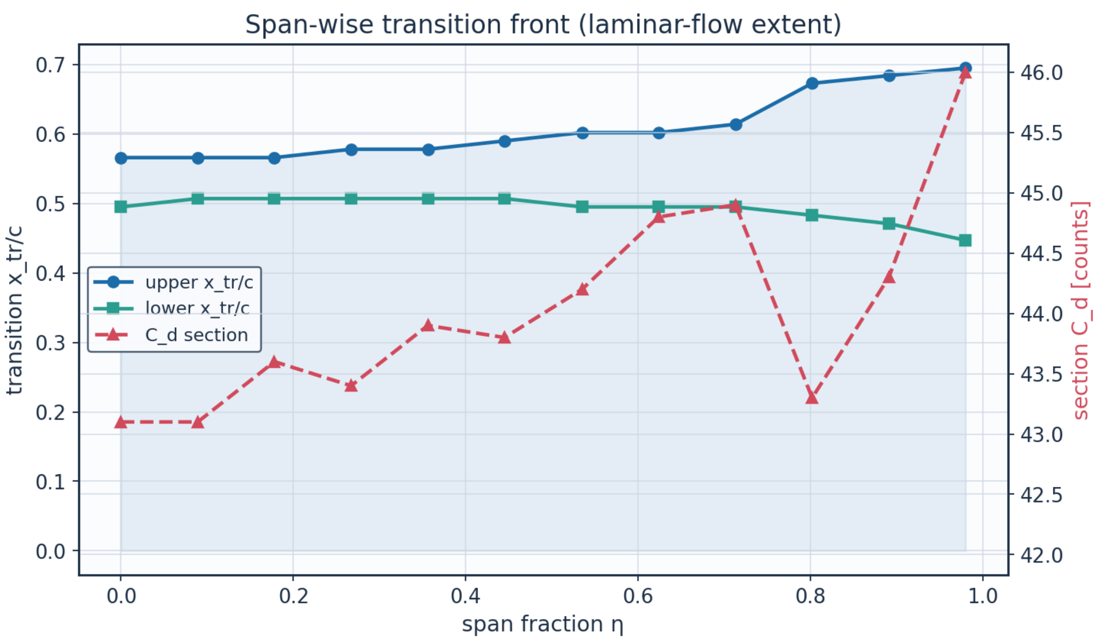


Fig. 28. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

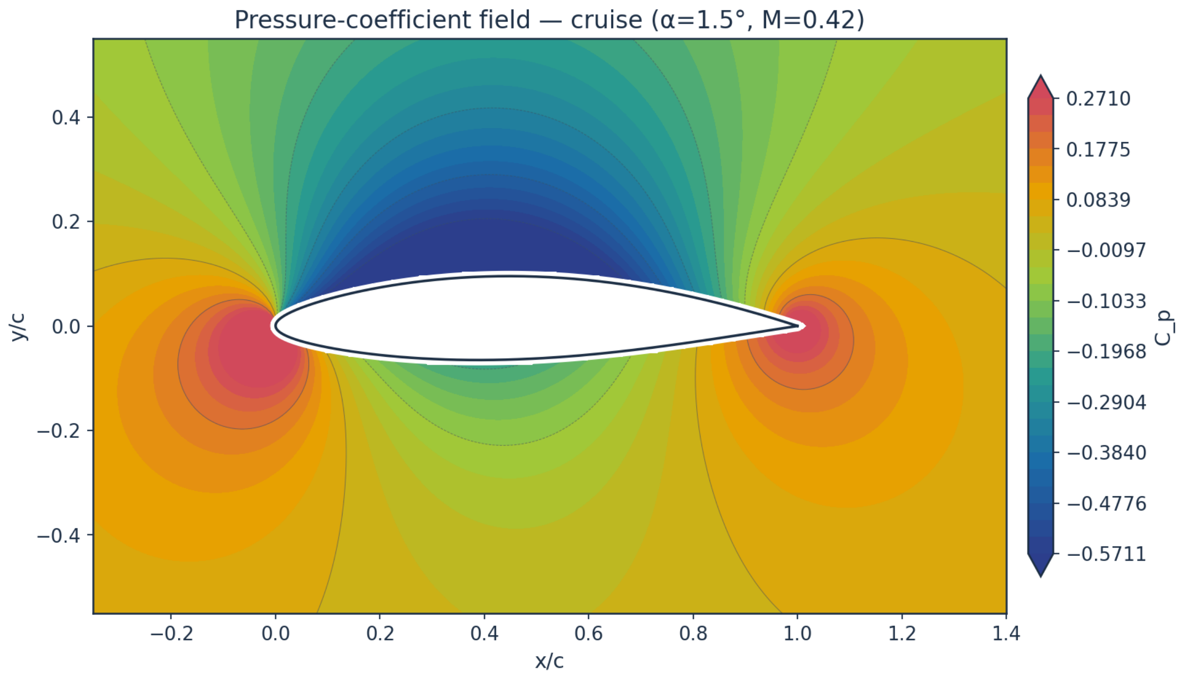


Fig. 29. Pressure-coefficient contour — cruise.

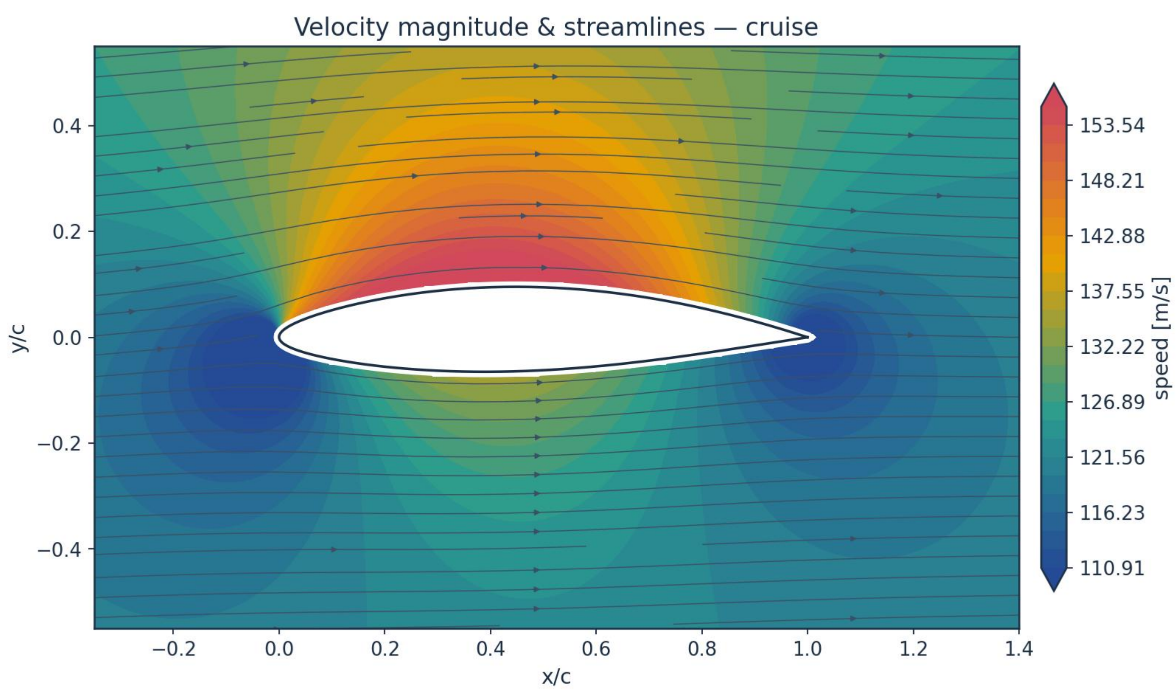


Fig. 30. Velocity magnitude and streamlines — cruise.

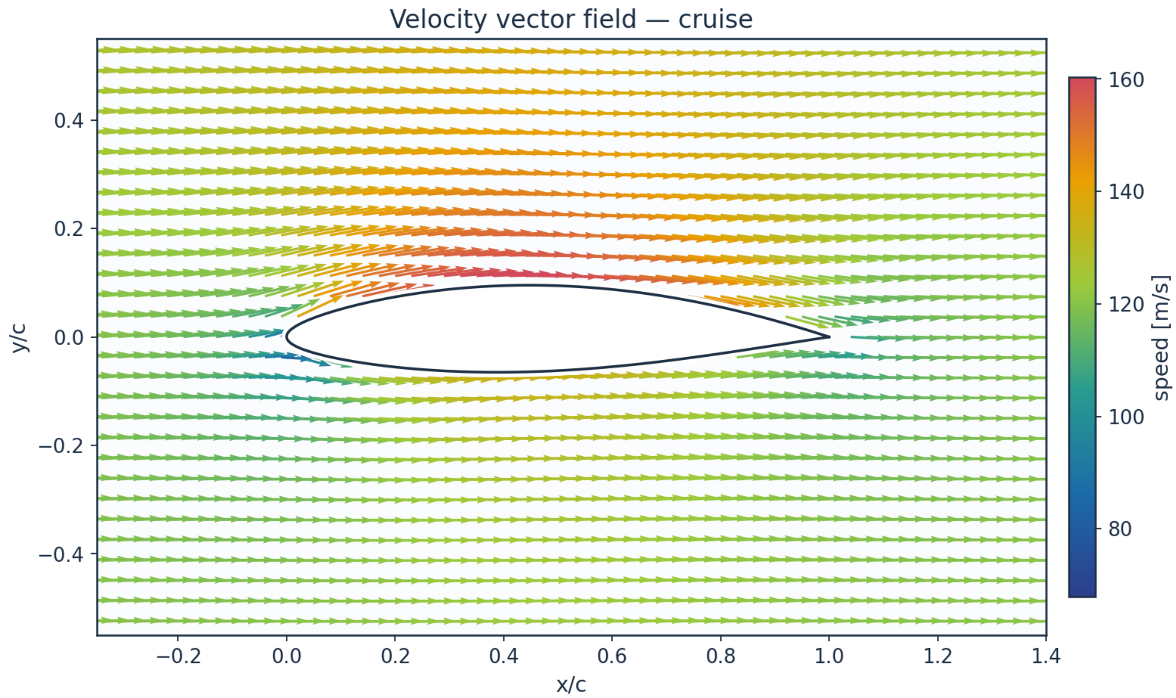


Fig. 31. Velocity vector field — cruise.

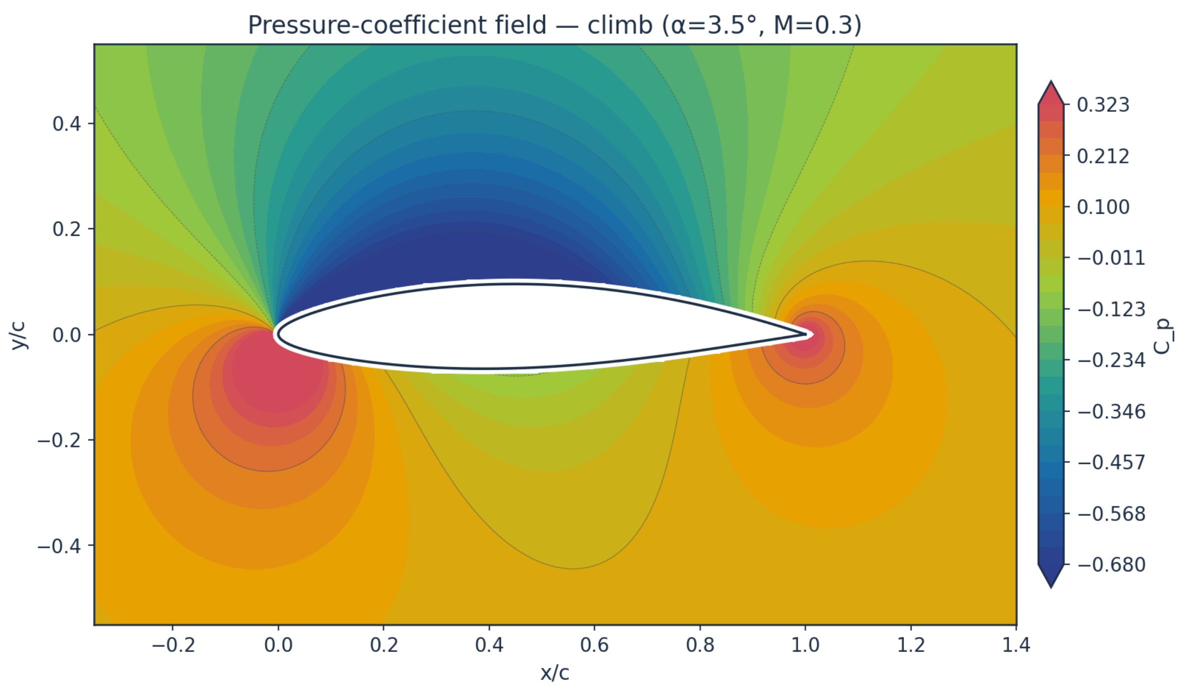


Fig. 32. Pressure-coefficient contour — climb.

10.2 Boundary-layer velocity and temperature profiles

The profiles are reconstructed from the marched state and not from an assumed shape. The laminar leg is the Falkner-Skan profile at the shape factor the march solved for, read from the same family that supplies the closure functions; the turbulent leg is the power law that shape factor implies, $H = (n+2)/n$; and the two are blended by the same intermittency that blends C_f, θ and H, each on its own thickness — $\delta_{99} = \eta_{99}\,\theta/\theta_\eta$ for the similarity profile and $\delta = \theta(n+1)(n+2)/n$ for the power law. Temperature profiles then follow from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

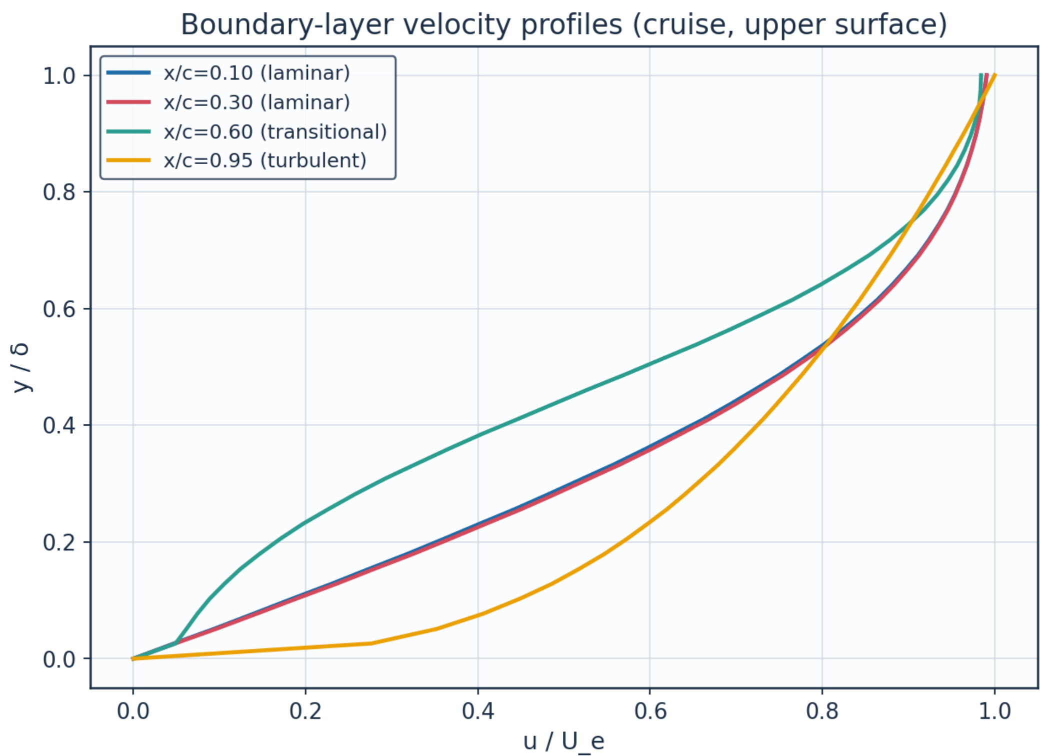


Fig. 33. Boundary-layer velocity profiles.

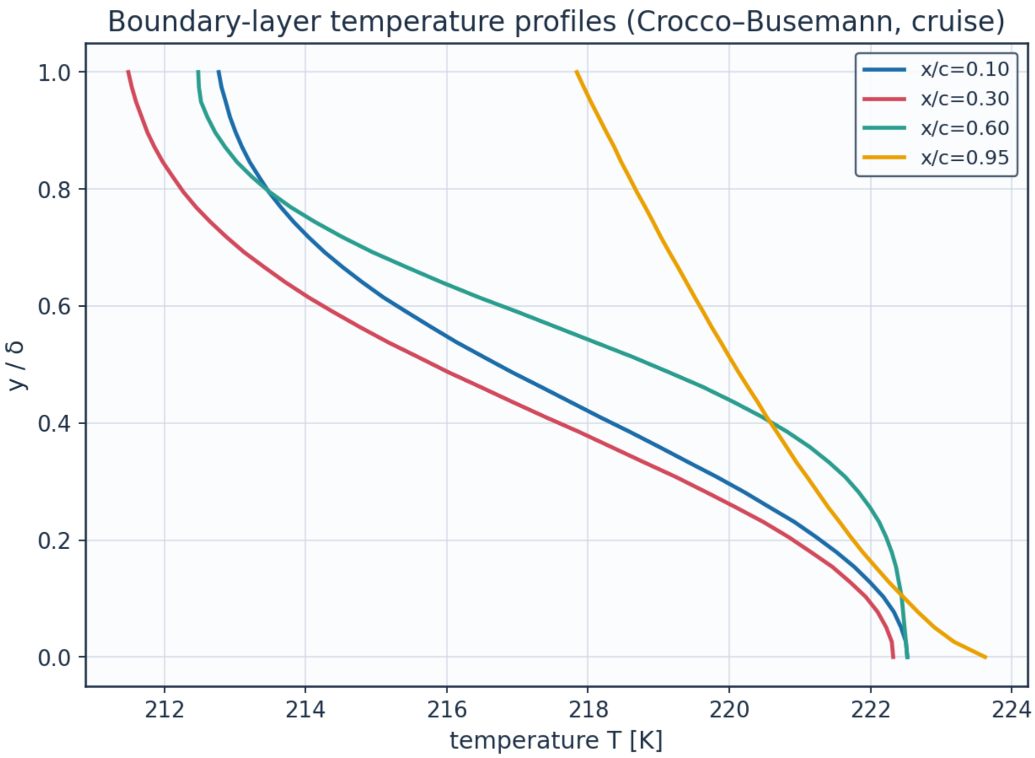


Fig. 34. Boundary-layer temperature profiles (K).

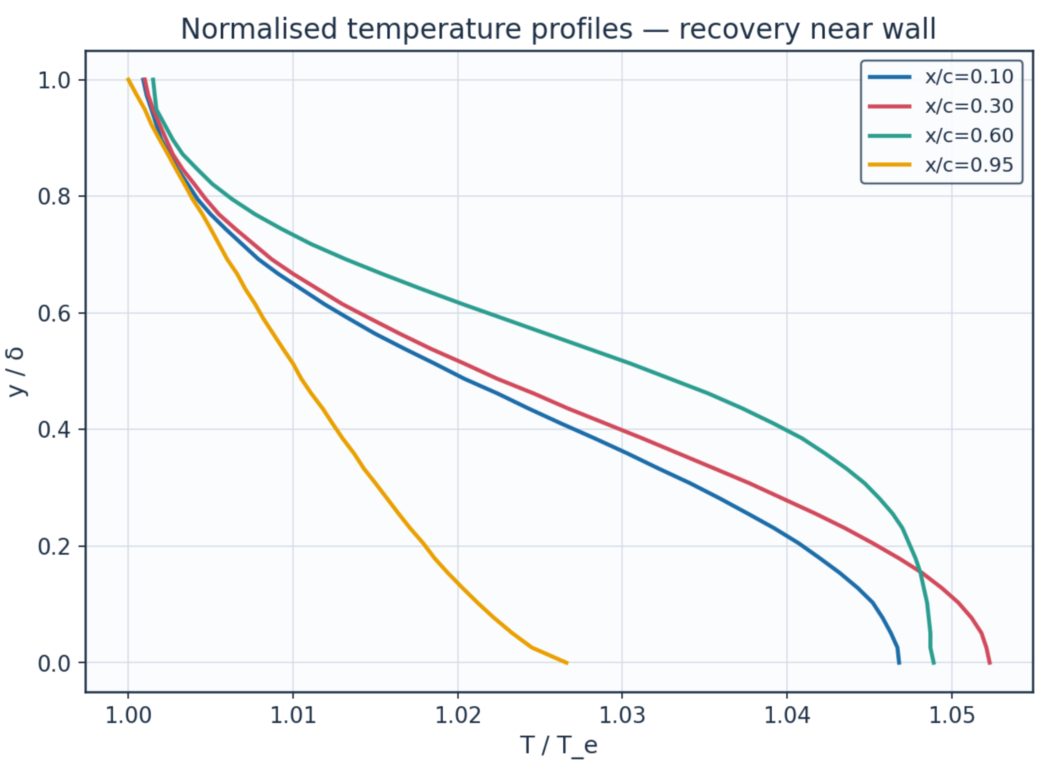


Fig. 35. Normalised temperature profiles T/T_e.

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K
x/c=0.10	0.1	0.0	0.0	0.0	1.0468	222.52
x/c=0.10	0.1	0.128	0.1422	0.2296	1.0443	221.99
x/c=0.10	0.1	0.256	0.2843	0.4424	1.0376	220.57
x/c=0.10	0.1	0.385	0.4265	0.6284	1.0283	218.59
x/c=0.10	0.1	0.513	0.5687	0.7766	1.0186	216.52
x/c=0.10	0.1	0.641	0.7109	0.8811	1.0105	214.8
x/c=0.10	0.1	0.769	0.853	0.9449	1.005	213.65
x/c=0.10	0.1	0.897	0.9952	0.978	1.002	213.01
x/c=0.30	0.3	0.0	0.0	0.0	1.0523	222.32
x/c=0.30	0.3	0.128	0.2341	0.2352	1.0494	221.71
x/c=0.30	0.3	0.256	0.4682	0.45	1.0417	220.09
x/c=0.30	0.3	0.385	0.7022	0.6352	1.0312	217.87
x/c=0.30	0.3	0.513	0.9363	0.7811	1.0204	215.59
x/c=0.30	0.3	0.641	1.1704	0.8833	1.0115	213.71
x/c=0.30	0.3	0.769	1.4045	0.9456	1.0055	212.45
x/c=0.30	0.3	0.897	1.6385	0.978	1.0023	211.76
x/c=0.60	0.6	0.0	0.0	0.0	1.0489	222.52
x/c=0.60	0.6	0.128	0.4454	0.1055	1.0483	222.4
x/c=0.60	0.6	0.256	0.8907	0.226	1.0464	221.99
x/c=0.60	0.6	0.385	1.3361	0.4045	1.0409	220.82
x/c=0.60	0.6	0.513	1.7815	0.6125	1.0305	218.63
x/c=0.60	0.6	0.641	2.2269	0.7978	1.0178	215.92
x/c=0.60	0.6	0.769	2.6722	0.9177	1.0077	213.79
x/c=0.60	0.6	0.897	3.1176	0.9721	1.0027	212.72
x/c=0.95	0.95	0.0	0.0	0.0	1.0266	223.62
x/c=0.95	0.95	0.128	3.1704	0.4856	1.0203	222.26
x/c=0.95	0.95	0.256	6.3408	0.6194	1.0164	221.4
x/c=0.95	0.95	0.385	9.5112	0.7144	1.013	220.67

(columns 2-7 of 13; station repeated on each block)

station	Me_edge	H_shape	recovery_r	delta_mm	intermittency_gamma	state
x/c=0.10	0.525	2.512	0.8485	1.1089	0.0	laminar
x/c=0.10	0.525	2.512	0.8485	1.1089	0.0	laminar
x/c=0.10	0.525	2.512	0.8485	1.1089	0.0	laminar
x/c=0.10	0.525	2.512	0.8485	1.1089	0.0	laminar
x/c=0.10	0.525	2.512	0.8485	1.1089	0.0	laminar
x/c=0.10	0.525	2.512	0.8485	1.1089	0.0	laminar
x/c=0.10	0.525	2.512	0.8485	1.1089	0.0	laminar
x/c=0.10	0.525	2.512	0.8485	1.1089	0.0	laminar
x/c=0.30	0.555	2.49	0.8485	1.8258	0.0	laminar
x/c=0.30	0.555	2.49	0.8485	1.8258	0.0	laminar
x/c=0.30	0.555	2.49	0.8485	1.8258	0.0	laminar
x/c=0.30	0.555	2.49	0.8485	1.8258	0.0	laminar
x/c=0.30	0.555	2.49	0.8485	1.8258	0.0	laminar
x/c=0.30	0.555	2.49	0.8485	1.8258	0.0	laminar
x/c=0.30	0.555	2.49	0.8485	1.8258	0.0	laminar
x/c=0.30	0.555	2.49	0.8485	1.8258	0.0	laminar
x/c=0.60	0.535	3.709	0.8539	3.4739	0.113	transitional
x/c=0.60	0.535	3.709	0.8539	3.4739	0.113	transitional
x/c=0.60	0.535	3.709	0.8539	3.4739	0.113	transitional
x/c=0.60	0.535	3.709	0.8539	3.4739	0.113	transitional
x/c=0.60	0.535	3.709	0.8539	3.4739	0.113	transitional
x/c=0.60	0.535	3.709	0.8539	3.4739	0.113	transitional
x/c=0.60	0.535	3.709	0.8539	3.4739	0.113	transitional
x/c=0.60	0.535	3.709	0.8539	3.4739	0.113	transitional

x/c=0.95	0.385	1.708	0.8961	24.7292	0.997	turbulent
x/c=0.95	0.385	1.708	0.8961	24.7292	0.997	turbulent
x/c=0.95	0.385	1.708	0.8961	24.7292	0.997	turbulent
x/c=0.95	0.385	1.708	0.8961	24.7292	0.997	turbulent

Table 20. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled). (columns 8-13 of 13; the table is split across 2 blocks so that no cell is compressed to illegibility, with station repeated on each)

(table sampled to 28 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

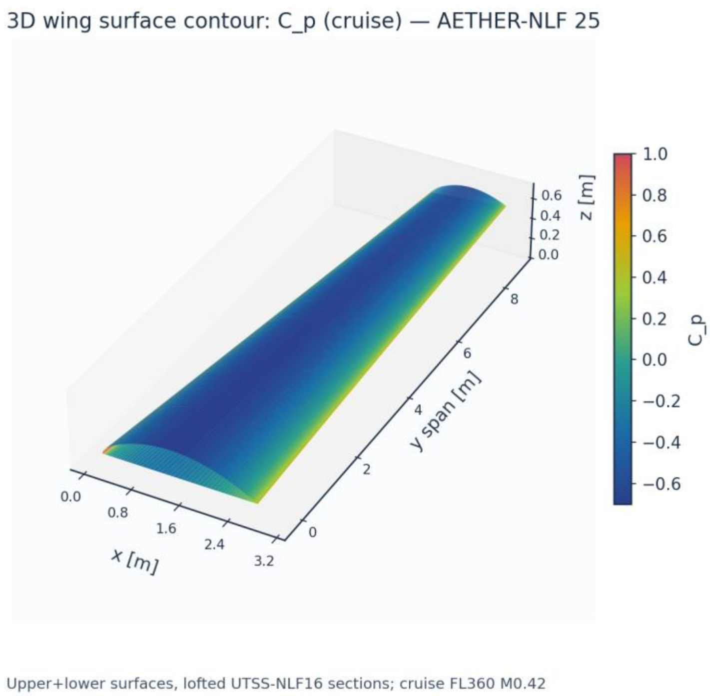


Fig. 36. 3-D wing surface contour — pressure C_p.

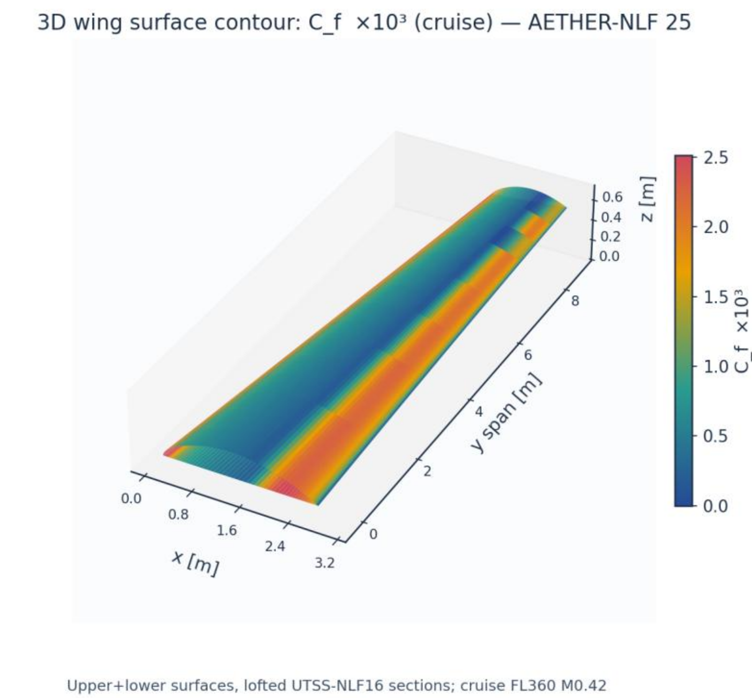


Fig. 37. 3-D wing surface contour — skin friction C_f (×10³).

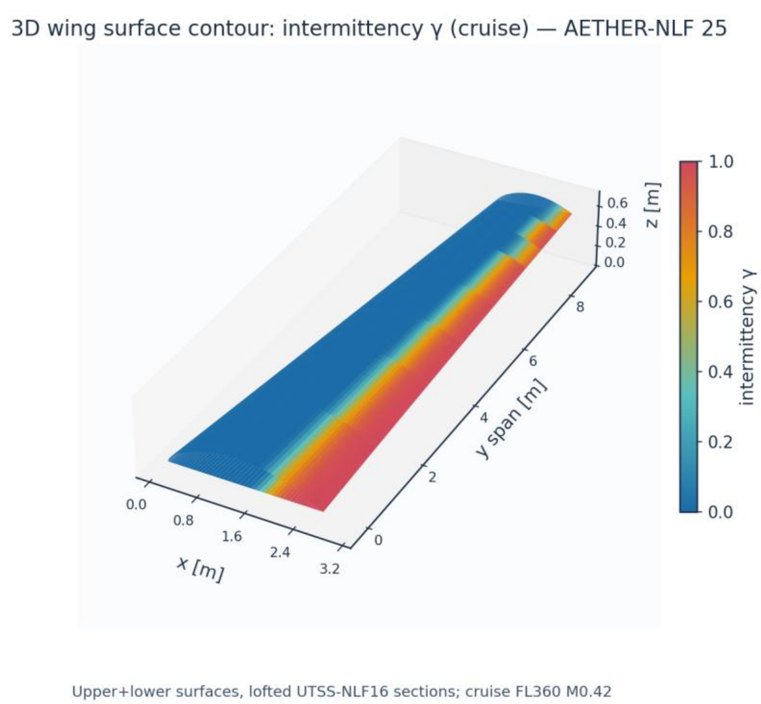


Fig. 38. 3-D wing surface contour — intermittency γ (transition front).

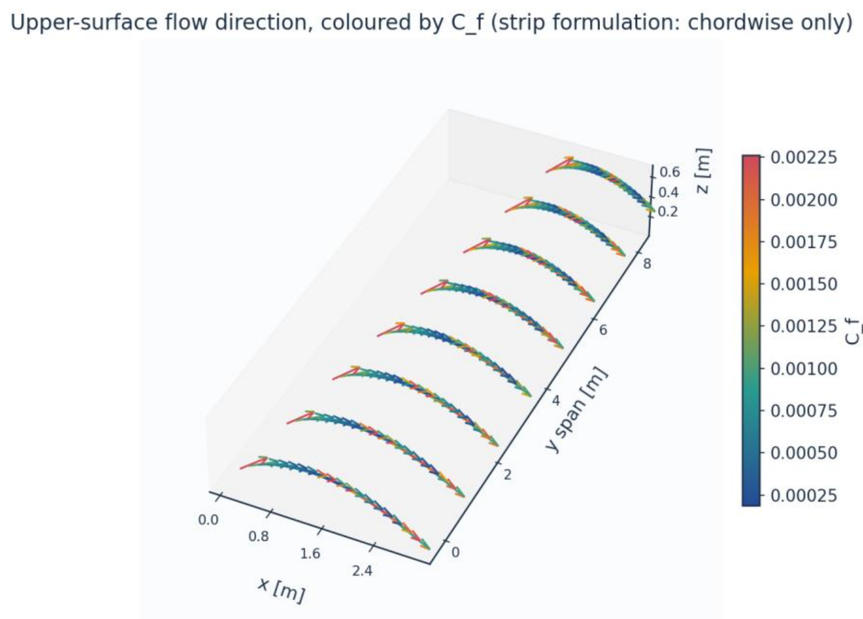


Fig. 39. Upper-surface flow direction coloured by C_f (chordwise only — the strip formulation carries no span-wise wall shear).

11. Validation and Calibration

To qualify as universal, the solver is validated against every published dataset the kernel's four branches reach — five flat plates spanning 0.03 to 6 per cent free-stream turbulence (bypass, natural and separation-induced transition), 86 aerofoil conditions on the NLF(1)-0416 section, and two swept wings from different facilities and eras — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number Re_θt is the metric on the plates and the transition location x_tr/c on the aerofoil and the swept wings. The skin-friction error is reported separately over the laminar run and over the turbulent run, on the stations where the measurement and the prediction are in the same state; pooled across transition it measures the onset error a second time, in the wrong units, because a plate whose onset is early by 16 % is then charged with the whole laminar-to-turbulent step in C_f over the interval between the two onsets.

case	Tu_inlet_pct	L_turb_mm	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_theta_t_exp_lo
ERCOFTAC T3A flat plate (bypass transition)	3.043	1.53	272.3	282.1	3.6	
ERCOFTAC T3A- flat plate (low-Tu	0.874	0.98	818.8	683.6	-16.5	

bypass)						
ERCOFTAC T3B flat plate (high-Tu bypass)	5.952	3.83	181.3	168.0	-7.3	
ERCOFTAC T3C4 flat plate (laminar separation bubble)	2.11		381.3	265.5	-30.4	309.3
Schubauer & Skramstad flat plate (natural transition)	0.03		1100.0	1100.7	0.1	

(columns 2-7 of 18; case repeated on each block)

case	Re_theta_t_exp_hi	Re_theta_t_err_bracket_pct	Re_x_tr_exp	Re_x_tr_pred	Cf_err_laminar_pct	n_pts_laminar
ERCOFTAC T3A flat plate (bypass transition)			1.348e+05	1.804e+05	3.8	4
ERCOFTAC T3A- flat plate (low-Tu bypass)			1.443e+06	1.060e+06	4.2	8
ERCOFTAC T3B flat plate (high-Tu bypass)			5.910e+04	6.405e+04	8.2	3
ERCOFTAC T3C4 flat plate (laminar separation bubble)	381.3	-14.2	1.785e+05	1.662e+05	21.9	9
Schubauer & Skramstad flat plate (natural transition)			2.800e+06	2.747e+06		0

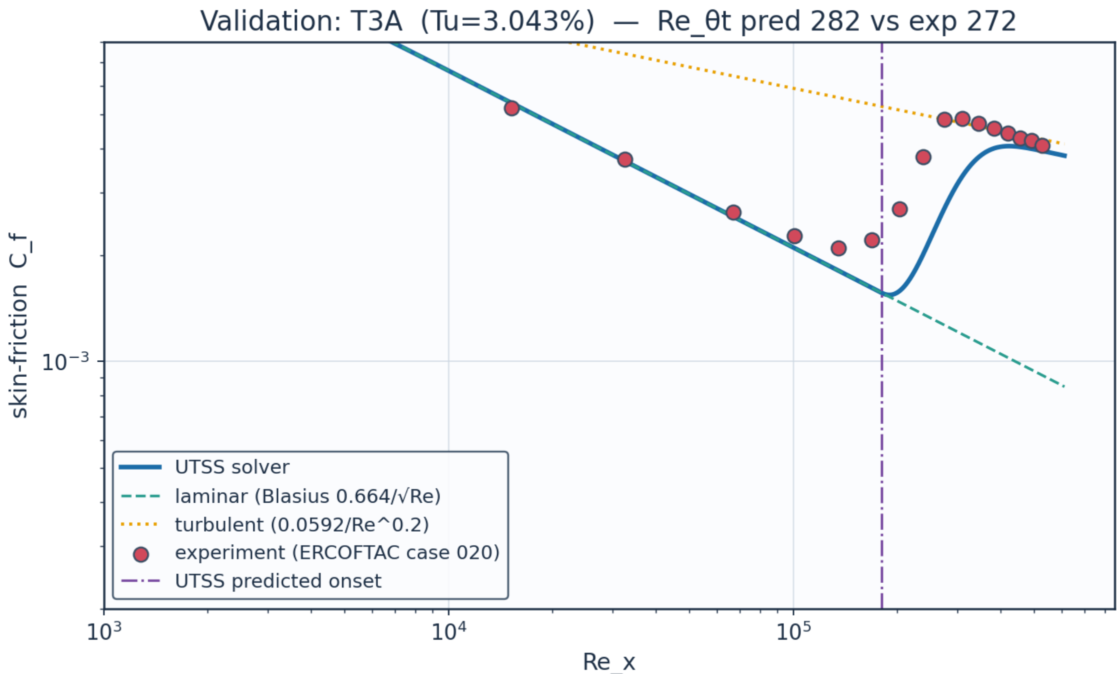
(columns 8-13 of 18; case repeated on each block)

case	Cf_err_turbulent_pct	n_pts_turbulent	Cf_err_all_pts_pct	Re_theta_meas_over_blasius	mechanism
ERCOFTAC T3A flat plate (bypass transition)	4.8	3	17.2	1.117	bypass
ERCOFTAC T3A- flat plate (low-Tu bypass)		0	121.6	1.027	bypass
ERCOFTAC T3B flat plate (high-Tu bypass)	8.2	9	12.4	1.123	bypass
ERCOFTAC T3C4 flat plate (laminar separation bubble)		0	34.9	1.359	separation

Schubauer & Skramstad flat plate (natural transition)		0			TS-natural

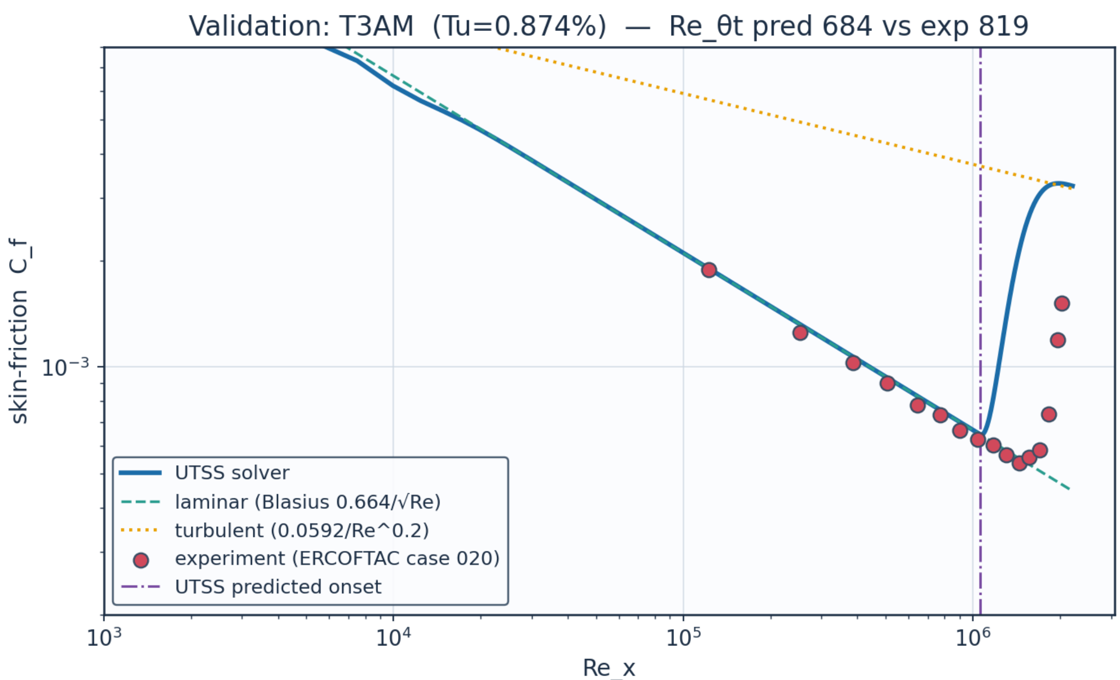
Table 21. Validation summary — predicted vs published Re_ϑt. (columns 14-18 of 18; the table is split across 3 blocks so that no cell is compressed to illegibility, with case repeated on each)

11.1 Flat plates



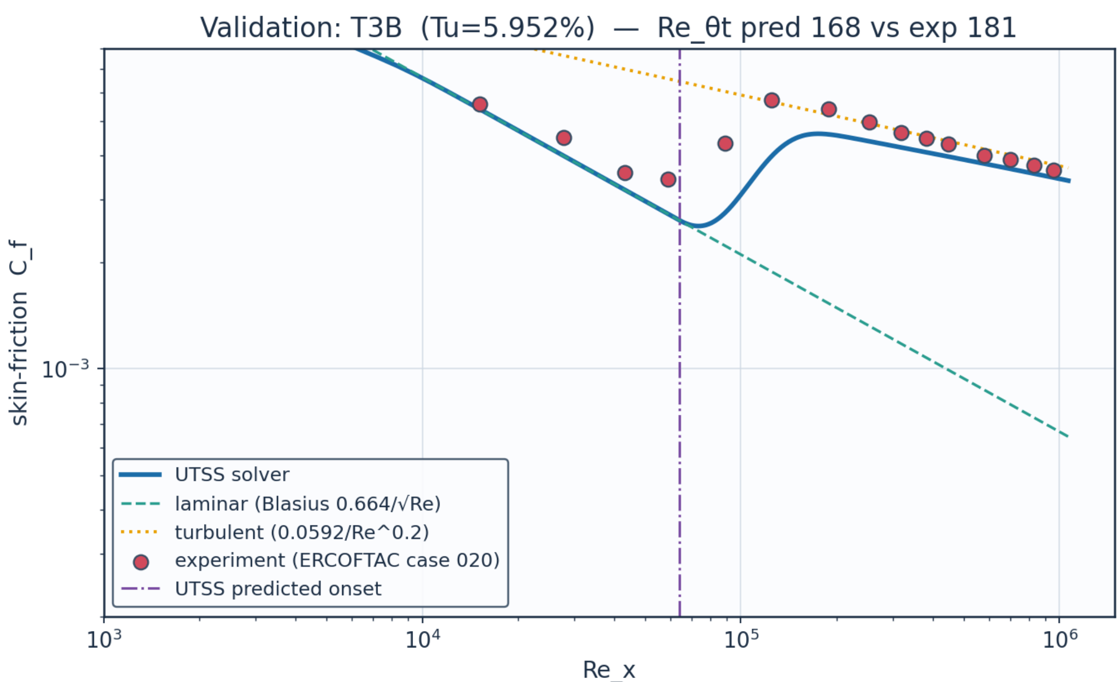
Source: Roach & Brierley (1990), ERCOFTAC T3A; Savill (1993); Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 40. Validation — ERCOFTAC T3A flat plate (Tu = 3.0 %, bypass).



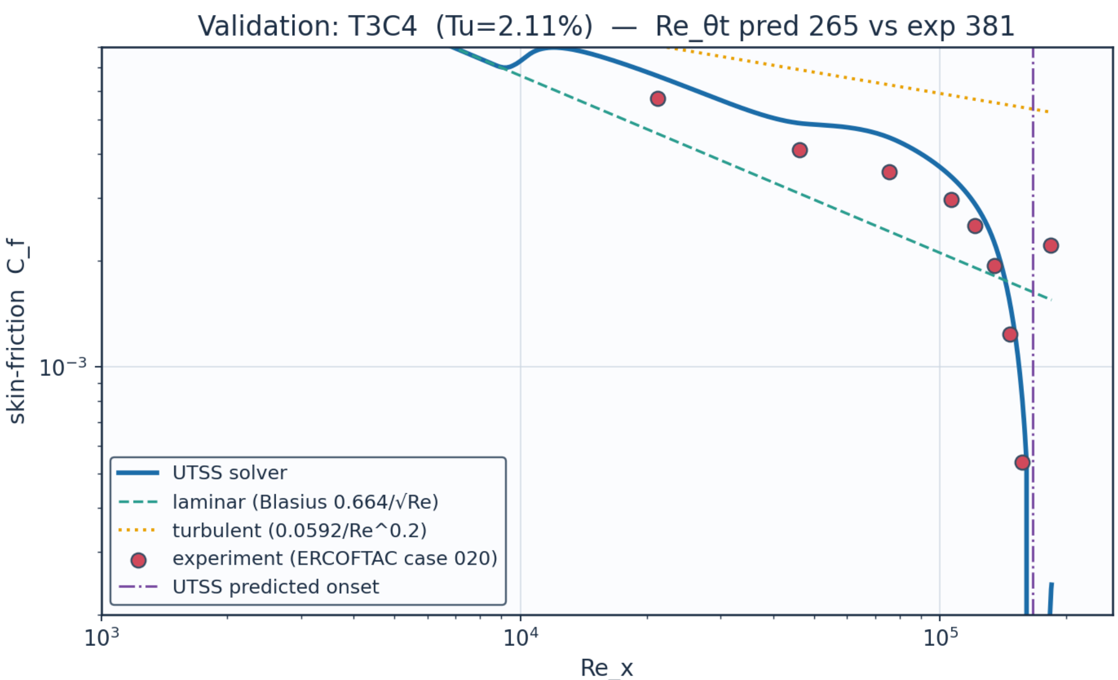
Source: Roach & Brierley (1990), ERCOFTAC T3A-; ERCOFTAC Classic Collection Case 020....

Fig. 41. Validation — ERCOFTAC T3A⁻ flat plate (Tu = 0.87 %, bypass).



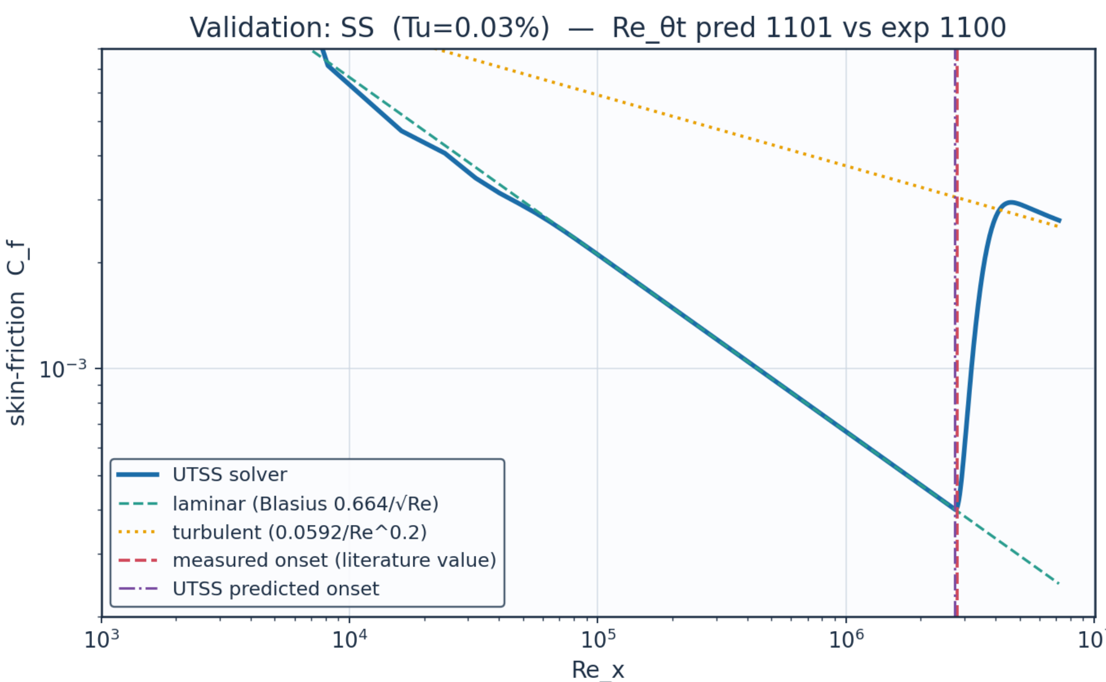
Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 42. Validation — ERCOFTAC T3B flat plate (Tu = 6.0 %, bypass).



Source: Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020, variable-pressure-grad...

Fig. 43. Validation — ERCOFTAC T3C4 flat plate (laminar separation bubble).



Source: Schubauer & Skramstad (1948), NACA Report 909....

Fig. 44. Validation — Schubauer & Skramstad natural transition (Tu = 0.03 %).

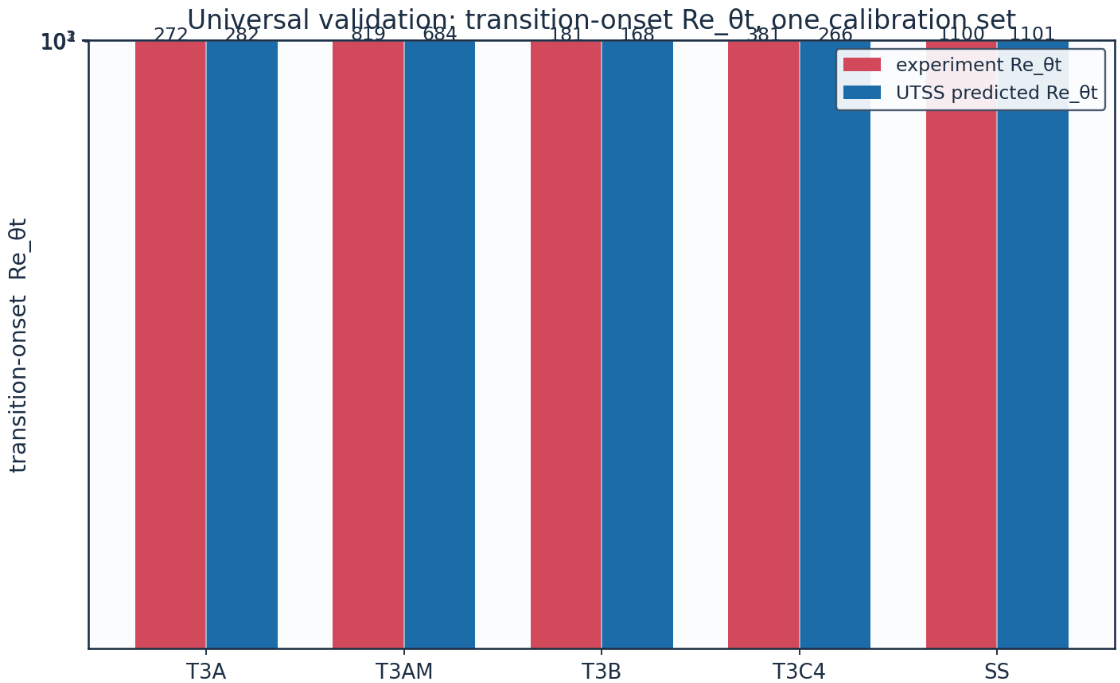


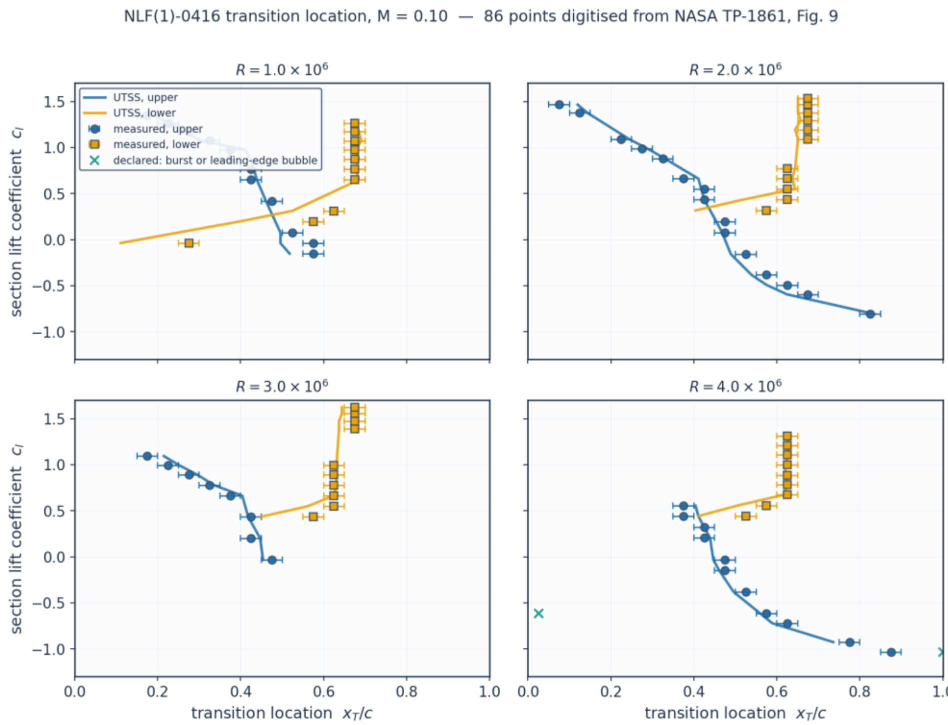
Fig. 45. Universal validation — Re_ϑt across all five plates.

11.2 NLF(1)-0416 aerofoil — 86 transition locations

The largest single body of evidence in this work, and genuinely out of sample. That was not always true: the anchor of the natural branch — the one quantity that branch takes from measurement — used to be chosen at whatever value put the most of these 86 predictions inside the experimental bracket, which made this a calibration set while this section, the figure below and the project README all called it otherwise. It is now set on the Schubauer & Skramstad plate alone, which it reproduces to 0.07 %, and the cost of that is reported rather than absorbed: the mean absolute error here rises from 0.0319c to 0.0334c and the bracket count falls from 51 to 50, which is what happens when a constant stops being fitted to the set it is scored on. Transition locations were digitised from Fig. 9 of the source report at four chord Reynolds numbers on both surfaces, and each condition is matched by trimming the incidence to the measured lift coefficient. The experiment brackets transition between adjacent orifices 0.05c apart, so its own uncertainty is ±0.025c and a prediction inside that band cannot be distinguished from the measurement. Both the natural (TS) and the separation-induced branches are selected on this set.

set	points	mean_abs_err_c	bias_c	within_bracket	within_bracket_pct
Upper surface	46	0.0385	-0.0113	24	52.2
Lower surface	40	0.0384	-0.0316	26	65.0
All	86	0.0384	-0.0207	50	58.1
Conditions the method accepts	84	0.0313	-0.0162	50	59.5

Table 22. NLF(1)-0416 error statistics by surface (aerofoil_nlf0416_summary.csv).



Bars show the +/-0.025c orifice-pitch bracket within which the experiment localises transition. Nothing is calibrated on this set: the one constant of the natural branch is set on the Schubauer & Skramstad plate alone.

Fig. 46. Transition location against lift coefficient at four chord Reynolds numbers, measurement bars = the ±0.025c orifice bracket.

Re_c	surface	c_l_exp	alpha_deg	c_l_solver	x_tr_c_exp	x_tr_c_pred
1.0e+06	upper	1.35	6.483	1.3503	0.175	0.192
1.0e+06	upper	1.174	5.036	1.1743	0.275	0.264
1.0e+06	upper	0.976	3.413	0.9762	0.375	0.411
1.0e+06	upper	0.765	1.687	0.765	0.425	0.432
1.0e+06	upper	0.419	-1.132	0.4187	0.475	0.461
1.0e+06	upper	-0.039	-4.846	-0.0387	0.575	0.496
1.0e+06	lower	1.262	5.759	1.2623	0.675	0.678
1.0e+06	lower	1.075	4.224	1.0753	0.675	0.691
1.0e+06	lower	0.874	2.578	0.8741	0.675	0.685
1.0e+06	lower	0.651	0.757	0.6508	0.675	0.685
1.0e+06	lower	0.197	-2.935	0.1968	0.575	0.397
2.0e+06	upper	1.466	7.438	1.4661	0.075	0.12
2.0e+06	upper	1.089	4.339	1.0893	0.225	0.245
2.0e+06	upper	0.883	2.652	0.8831	0.325	0.329
2.0e+06	upper	0.551	-0.058	0.5507	0.425	0.418
2.0e+06	upper	0.198	-2.927	0.1978	0.475	0.454
2.0e+06	upper	-0.156	-5.791	-0.1552	0.525	0.489
2.0e+06	upper	-0.491	-8.514	-0.491	0.625	0.575
2.0e+06	upper	-0.807	-11.077	-0.8069	0.825	0.835
2.0e+06	lower	1.469	7.463	1.4691	0.675	0.651
2.0e+06	lower	1.294	6.022	1.2943	0.675	0.658
2.0e+06	lower	1.093	4.372	1.0933	0.675	0.651
2.0e+06	lower	0.663	0.855	0.6628	0.625	0.637
2.0e+06	lower	0.434	-1.01	0.4337	0.625	0.518
3.0e+06	upper	1.094	4.38	1.0943	0.175	0.215
3.0e+06	upper	0.89	2.709	0.8901	0.275	0.296
3.0e+06	upper	0.662	0.847	0.6618	0.375	0.404

3.0e+06	upper	0.202	-2.894	0.2018	0.425	0.447
3.0e+06	lower	1.623	8.733	1.6227	0.675	0.644
3.0e+06	lower	1.472	7.487	1.4721	0.675	0.637

(columns 2-7 of 15; Re_c repeated on each block)

Re_c	err_c	err_pct	within_bracket	degenerate	burst	le_sep
1.0e+06	0.017	9.6	True	False	False	False
1.0e+06	-0.011	-4.2	True	False	False	False
1.0e+06	0.036	9.7	False	False	False	False
1.0e+06	0.007	1.7	True	False	False	False
1.0e+06	-0.014	-3.0	True	False	False	False
1.0e+06	-0.079	-13.7	False	False	False	False
1.0e+06	0.003	0.5	True	False	False	False
1.0e+06	0.016	2.4	True	False	False	False
1.0e+06	0.01	1.4	True	False	False	False
1.0e+06	0.01	1.4	True	False	False	False
1.0e+06	-0.178	-30.9	False	False	False	False
2.0e+06	0.045	59.7	False	False	False	False
2.0e+06	0.02	8.8	True	False	False	False
2.0e+06	0.004	1.1	True	False	False	False
2.0e+06	-0.007	-1.6	True	False	False	False
2.0e+06	-0.021	-4.5	True	False	False	False
2.0e+06	-0.036	-6.8	False	False	False	False
2.0e+06	-0.05	-8.1	False	False	False	False
2.0e+06	0.01	1.3	True	False	False	False
2.0e+06	-0.024	-3.5	True	False	False	False
2.0e+06	-0.017	-2.5	True	False	False	False
2.0e+06	-0.024	-3.5	True	False	False	False
2.0e+06	0.012	2.0	True	False	False	False
2.0e+06	-0.107	-17.1	False	False	False	False
3.0e+06	0.04	22.8	False	False	False	False
3.0e+06	0.021	7.5	True	False	False	False
3.0e+06	0.029	7.8	False	False	False	False
3.0e+06	0.022	5.1	True	False	False	False
3.0e+06	-0.031	-4.6	False	False	False	False
3.0e+06	-0.038	-5.6	False	False	False	False

(columns 8-13 of 15; Re_c repeated on each block)

Re_c	x_sep_c	mechanism
1.0e+06		TS-natural
1.0e+06		TS-natural
1.0e+06	0.335	separation
1.0e+06	0.356	separation
1.0e+06	0.383	separation
1.0e+06		TS-natural
1.0e+06	0.582	separation
1.0e+06	0.582	separation
1.0e+06	0.575	separation
1.0e+06	0.568	separation
1.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06		TS-natural

2.0e+06		TS-natural
2.0e+06	0.369	separation
2.0e+06	0.404	separation
2.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06	0.589	separation
2.0e+06	0.589	separation
2.0e+06	0.582	separation
2.0e+06	0.568	separation
2.0e+06		TS-natural
3.0e+06		TS-natural
3.0e+06		TS-natural
3.0e+06	0.363	separation
3.0e+06	0.404	separation
3.0e+06	0.596	separation
3.0e+06	0.589	separation

Table 23. NLF(1)-0416 point-by-point comparison (aerofoil_nlf0416.csv, sampled). (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with Re_c repeated on each)

(table sampled to 30 rows; full data in 06_validation/aerofoil_nlf0416.csv)

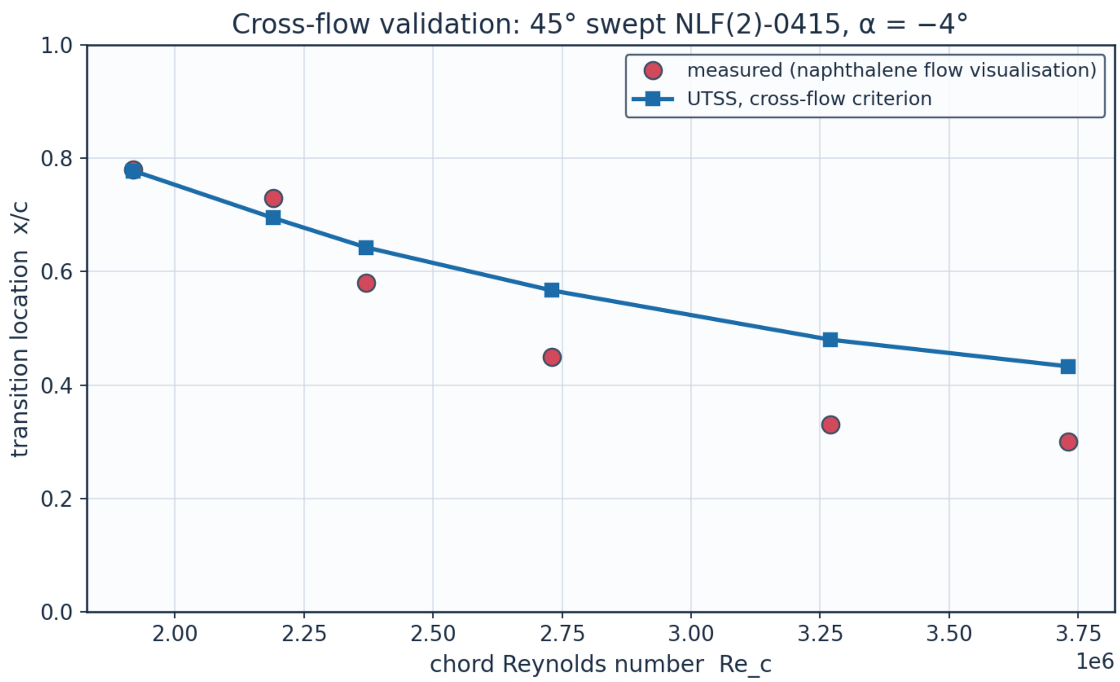
11.3 Swept wings — the cross-flow branch

The cross-flow coefficient is set on the first of these two experiments and nothing is calibrated on the second, which is a different facility, section and era. The branch reproduces the calibration set to 14.7 % at the frozen constant C1 = 150. On the independent set that same constant gives 51.7 %, and C1 = 200 gives 18.4 % — so the criterion has the right functional form on both wings but not one critical value that serves both. The independent set is therefore tabulated at BOTH ends of that band: reporting only C1 = 150 understates what the criterion does here, and reporting only C1 = 200 would be a per-case re-tune of the kind this work is claiming not to need. The spread between the two columns is the limitation, stated rather than averaged away. Nothing else in the model differs between the two columns.

What each experiment requires of the criterion is measured rather than asserted (Table 28). The march is run with every branch disabled so that it reaches the measured transition station, and the criterion is evaluated there. Dagenhart & Saric require a critical Re_θ2 of 153 with a 18.0 % coefficient of variation over six chord Reynolds numbers; Boltz et al. require 234 with 4.0 % over four sweep angles and a factor of three in chord Reynolds number. Each facility is therefore internally consistent — the second markedly so — and the two differ by 53 %. That is the shape of a receptivity difference rather than of a criterion with the wrong form: stationary cross-flow vortices are seeded by leading-edge roughness, and neither report documents the surface finish. THREE attempts to close the gap fail and are recorded rather than dropped. Solving the swept sections in the plane normal to the leading edge, which is the correct mean flow and had not been done, widens the gap rather than closing it — it was the strongest remaining physical candidate and it is now tested rather than argued about. Replacing the constant surrogate by the exact Falkner-Skan-Cooke factor K(λ) makes matters worse, taking the pooled coefficient of variation from 24 to 77 % and inverting the ratio between the two sets. Giving the cross-flow branch its own amplification threshold, separate from the one Mack's relation supplies for Tollmien-Schlichting waves — which is defensible, since a stationary cross-flow vortex is not seeded by free-stream turbulence — moves the independent set only from 55 to 51 % as that threshold is taken from N = 2 to N = 12, with C1 refitted on the calibration set at each value. The reason is that once Re_θ2 exceeds C1 the amplification builds so quickly that the threshold is nearly redundant with C1 itself, so the two constants cannot be separated by these data.

Re_c	x_tr_c_exp	x_tr_c_pred	err_pct	Re_theta_at_onset	mechanism
1.920e+06	0.78	0.778	-0.3	577.5	separation
2.190e+06	0.73	0.695	-4.8	475.0	crossflow
2.370e+06	0.58	0.643	10.9	471.4	crossflow
2.730e+06	0.45	0.567	25.9	474.3	crossflow
3.270e+06	0.33	0.48	45.6	474.7	crossflow
3.730e+06	0.3	0.433	44.5	478.6	crossflow

Table 24. Cross-flow validation, 45° swept NLF(2)-0415 (Dagenhart & Saric — the calibration set).



Source: Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wi...

Fig. 47. Transition location against chord Reynolds number, 45° swept NLF(2)-0415.

sweep_deg	alpha_deg	Re_c	x_tr_c_exp	x_tr_c_pred_C1_150	err_pct_C1_150	mechanism
20.0	0.0	2.700e+07	0.207	0.093	-55.0	crossflow
30.0	0.0	1.500e+07	0.213	0.093	-56.2	crossflow
40.0	0.0	1.200e+07	0.2	0.098	-51.1	crossflow
50.0	0.0	9.500e+06	0.211	0.122	-42.0	crossflow

(columns 2-7 of 9; sweep_deg repeated on each block)

sweep_deg	x_tr_c_pred_C1_200	err_pct_C1_200
20.0	0.161	-22.4
30.0	0.161	-24.6
40.0	0.161	-19.7
50.0	0.203	-3.8

Table 25. Independent swept-wing check, NACA 64(2)A015 (Boltz et al.) at both ends of the reported cross-flow band — nothing calibrated here. (columns 8-9 of 9; the table is split across 2 blocks so that no cell is compressed to illegibility, with sweep_deg repeated on each)

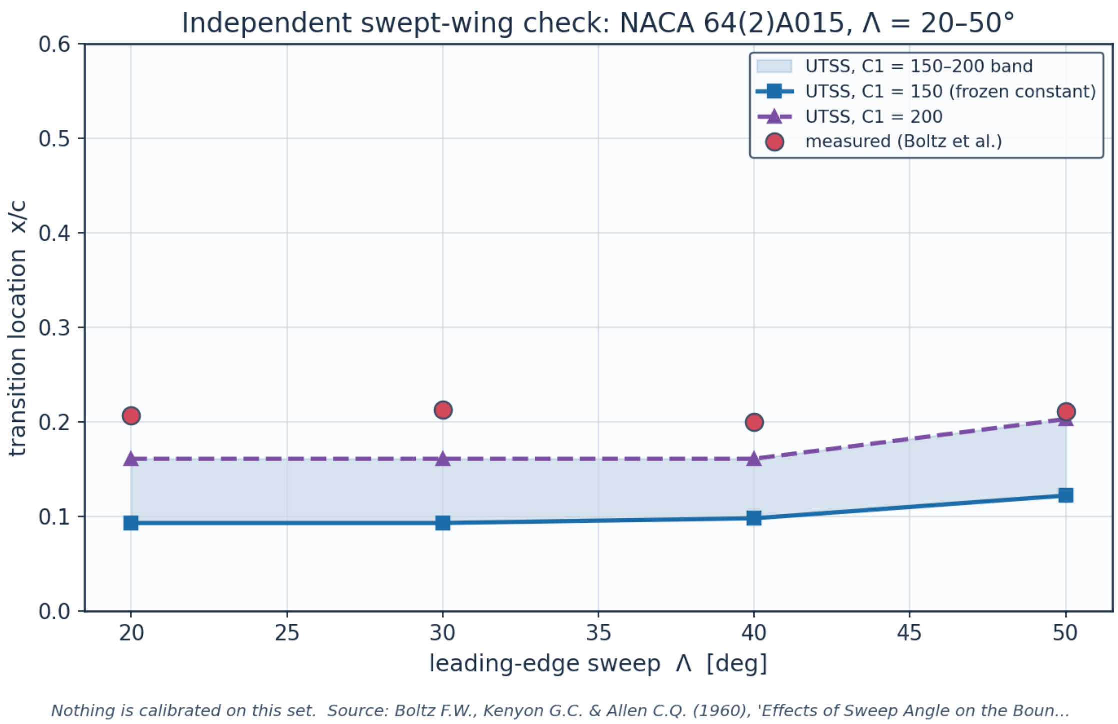


Fig. 48. Transition location against sweep angle, NACA 64(2)A015, with the C1 = 150–200 band shaded.

11.3a Where the flat-plate residuals come from

The residuals are accounted for here, and the accounting is generated by gen_validation.py rather than argued.

The laminar branch is not where they are. Compared against the model's own marched momentum thickness at the same stations — not against flat-plate Blasius, which is the wrong reference for the one plate that has a pressure gradient — the measured laminar layer agrees to within 6 % on all four plates that carry C_f data, and to 2 % on T3C4. An earlier version of this report attributed the T3C4 residual to the pre-transitional thickening a laminar layer undergoes in a turbulent free stream, on the strength of the measured momentum thickness being 1.36 times Blasius at onset. That comparison was wrong: a laminar layer in an adverse gradient is thicker than Blasius for a reason that has nothing to do with turbulence, and against the march that carries the gradient the agreement is 2 %.

The T3C4 residual is in the bubble, and Table 27 localises it — but not where this study previously placed it. Three things are now established. First, the shape-factor cap is gone: $H^*(H)$ folds at $H = 4.03$ where the attached and reverse-flow branches meet, so the inversion $H = H(H^*)$ is not unique there, but a march knows which branch it is on because it arrived continuously, and taking the root nearest the previous H carries it across. The reverse branch itself was stopped at $H = 4.99$ by the continuation parameter, which was a choice and not a limit — it continues smoothly to $H = 6.41$, past the measured 5.17.

Second, and this is the substantive point: THE MOMENTUM INTEGRAL CANNOT BE CLOSED WITH THIS EXPERIMENT'S OWN DATA. Across the measured bubble, reproducing the measured $d\theta/dx = 0.00591\ m^{-1}$ from the measured $dU\_e/dx = -0.200\ s^{-1}$ and $\theta = 2.70$ mm requires a shape factor of 19. The measurement itself reports 5.17, which returns $0.00202\ m^{-1}$ — short by a factor of 2.93. No integral method closed on any physical profile family reproduces this bubble, so the shape-factor cap was never the leading term, and the earlier attribution of a factor 1.32 of the shortfall to that cap was accounting for the wrong thing.

Third, onset is not resolved to a station here. It is taken as the station of minimum measured C_f, as on the other plates. On this plate C_f is 1.87×10^{-4} at $x = 1.295$ m and 1.83×10^{-4} at $x = 1.395$ m — two per cent apart, in a hot-film measurement of a quantity at its floor. They are not distinguishable, so onset is bracketed by them, Re_θ from 309.3 to 381.3, and quoting the second alone reports the end of the plateau as though it were the beginning. Against that bracket the error is −14.2 %, not −30.4 %. The point value is retained in the tables for continuity with the literature; the bracket is what the residual should be read against.

Reading the amplification rate at the marched shape factor instead of at the developed reverse-flow profile became possible once the march crossed the fold, and looks more principled because it removes a constant. It is rejected on measurement: the T3C4 bubble lengthens from 0.052 to 0.087 m, towards the measured 0.10, but the 86 NLF(1)-0416 conditions collapse from 50 to 21 inside the bracket. The reason is physical — the marched H is the integral shape factor of the whole dead-air region, still near 3.4, while the detached shear layer riding on it is inflectional from the moment the flow leaves the wall. It is exposed as cal["bub_sigma_local"] so the test is reproducible.

Conditioning. In a decaying stream the onset threshold rises while Re_θ grows only as the square root of distance, so the two curves close at a shallow angle and the crossing is sensitive. Shifting the threshold by ±10 % moves the predicted transition location by a factor of 3.5 on T3A, 2.5 on T3A⁻ and 2.0 on T3B. A correlation accurate to ten per cent cannot locate transition to ten per cent in such a flow.

case	branch	Re_theta_t_pred	Re_theta_t_exp	err_pct	laminar_run_meas_over_march	n_laminar_stations
ERCOFTAC T3A flat plate (bypass transition)	bypass	282.1	272.3	3.6	1.058	6
ERCOFTAC T3A- flat plate	bypass	683.6	818.8	-16.5	0.988	8

(low-Tu bypass)						
ERCOFTAC T3B flat plate (high-Tu bypass)	bypass	168.0	181.3	-7.3	1.058	4
ERCOFTAC T3C4 flat plate (laminar separation bubble)	separation	265.5	381.3	-30.4	1.02	9
Schubauer & Skramstad flat plate (natural transition)	TS-natural	1100.7	1100.0	0.1		0

(columns 2-7 of 8; case repeated on each block)

case	location_gain_bypass_threshold
ERCOFTAC T3A flat plate (bypass transition)	3.5
ERCOFTAC T3A- flat plate (low-Tu bypass)	2.5
ERCOFTAC T3B flat plate (high-Tu bypass)	2.0
ERCOFTAC T3C4 flat plate (laminar separation bubble)	
Schubauer & Skramstad flat plate (natural transition)	

Table 26. The laminar branch against the measurements, and the sensitivity of the crossing (residual_diagnostics.csv). (columns 8-8 of 8; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

quantity	model	measured	note
separation station x [m]	1.236	1.295	model / first station at the C_f floor
reattachment station x [m]	1.288	1.395	model / last station at the C_f floor
bubble length [m]	0.052	0.1	measured length is a lower bound
dtheta/dx across the bubble [1/m]	1.891e-03	5.906e-03	momentum integral is exact given U_e and H
shape factor at reattachment	3.39	5.17	model bounded by the attached Falkner-Skan branch, H <= 3.997
dU_e/dx over the bubble [1/s]	-0.3782	-0.38	smoothing spline against the tabulated points
max |U_e spline - tabulated| [m/s]	0.011	0.01	tabulated to 0.01 m/s, so the spline is within quotation

Table 27. The T3C4 bubble, modelled against measured (bubble_diagnostics.csv). The momentum integral across the dead-air region is exact given U_e and H, so these are the whole of the residual.

11.4 What the two swept-wing experiments require of the criterion

dataset	criterion	mean_critical_value	coeff_of_variation_pct	n_points
Dagenhart & Saric (calibration)	surrogate Re_theta2	153.1	18.0	6
Dagenhart & Saric (calibration)	exact Falkner-Skan-Cooke Re_cf	429.3	54.0	6
Boltz et al.	surrogate Re_theta2	233.8	4.0	4

(independent)				
Boltz et al. (independent)	exact Falkner-Skan-Cooke Re_cf	117.8	4.8	4
Both facilities pooled	surrogate Re_theta2	185.4	24.4	10
Both facilities pooled	exact Falkner-Skan-Cooke Re_cf	304.7	77.3	10

Table 28. What each swept-wing experiment requires of the cross-flow criterion, evaluated at the measured transition station (crossflow_criticals_summary.csv).

dataset	sweep_deg	Re_c	x_tr_c_measured	Re_theta	Re_theta2_surrogate	Re_cf_exact_FSC
Dagenhart & Saric (calibration)	45.0	1.920e+06	0.78	617.7	205.3	814.9
Dagenhart & Saric (calibration)	45.0	2.190e+06	0.73	513.4	170.6	677.3
Dagenhart & Saric (calibration)	45.0	2.370e+06	0.58	447.4	148.7	354.4
Dagenhart & Saric (calibration)	45.0	2.730e+06	0.45	418.2	139.0	281.7
Dagenhart & Saric (calibration)	45.0	3.270e+06	0.33	382.3	127.1	226.3
Dagenhart & Saric (calibration)	45.0	3.730e+06	0.3	385.6	128.2	221.1
Boltz et al. (independent)	20.0	2.700e+07	0.207	1476.1	237.3	119.8
Boltz et al. (independent)	30.0	1.500e+07	0.213	1029.7	242.0	119.6
Boltz et al. (independent)	40.0	1.200e+07	0.2	787.5	237.9	123.4
Boltz et al. (independent)	50.0	9.500e+06	0.211	605.1	217.9	108.4

Table 29. Point-by-point cross-flow criterion values at the measured transition stations (crossflow_criticals.csv).

The gap cannot be converted into a roughness ratio by this method, and saying why is firmer than attributing it to receptivity by elimination. For a roughness-seeded stationary vortex the natural currency is amplification, $\Delta N = \ln(A_{0,1}/A_{0,2})$, which is the ratio of initial amplitudes and so of effective leading-edge roughness. But the rate this branch integrates is read off a separated STREAMWISE profile and is nearly Reynolds-independent — 0.0417 at Re_θ = 200 against 0.0461 at 8000 — so N_cf is essentially σ times run length over θ, which scales as $\sqrt{Re\_c}$: a property of the chord Reynolds number, not of the cross-flow instability. The two facilities differ by a factor of seven in Re_c, and the N they require at their own measured stations is 0.0–11.9 and 43.1–129.2 respectively (Table 30), while the critical cross-flow Reynolds number is tight within each, at 19.7 and 4.6 per cent. The elimination behind the receptivity attribution was therefore incomplete: it had not considered that the branch's own rate carries no cross-flow physics, which is a defect of the model and not a property of the experiments.

What would close it is a defined piece of work on the method rather than a request for measurements on two wings from 1960 and 1999: the Orr-Sommerfeld problem solved on the Falkner-Skan-Cooke CROSS-FLOW profile — which stability.fsc_profile already returns — and tabulated the way the streamwise rates of Sec. 4.3 are. Both closures were compared over both facilities while this was established: the amplification integral gives 21.8 and 51.1 per cent, the local C1 criterion 17.2 and 55.1, pooled 33.5 against 32.4. Neither dominates, so the shipped form is kept and the comparison recorded rather than the choice asserted.

dataset	Re_theta2_mean	Re_theta2_cov_pct	N_cf_min	N_cf_max	n_points
Dagenhart & Saric (calibration)	153.1	19.7	0.0	11.92	6
Boltz et al. (independent)	233.8	4.6	43.14	129.19	4

Table 30. What each swept-wing facility requires, in the two currencies: a critical cross-flow Reynolds number, which is consistent within each facility, and an amplification factor, which is not comparable between them.

One explanation can be ruled out rather than merely doubted. If the difference between the two facilities were a Reynolds-number effect the criterion is missing, the requirement would have to vary with chord Reynolds number in the same direction within a facility as it does between them. It does not. Within Dagenhart & Saric the required critical value FALLS steeply with chord Reynolds number, by 251 per decade with a correlation of −0.88 over a factor of two in Re_c; Boltz et al. sit at six times that Reynolds number and require 42 % MORE, not less, and are themselves flat across a factor of three. The between-facility offset therefore has the opposite sign to the within-facility trend, and no monotone function of Re_c can carry one set into the other. That is what leaves receptivity — the leading-edge surface finish neither report documents — as the explanation, and it is now a measurement rather than an appeal to the literature.

Two numbers in this section have to be reconciled or they look inconsistent: the band is reported as C1 = 150–200, while Table 28 says the independent facility requires a critical Re_θ2 of 234. They are different quantities. C1 does not fire the branch; it starts the amplification integral, which then decides where transition is placed, so the EFFECTIVE critical value at the predicted station is higher than C1. Measured at the four Boltz conditions it is 157 for C1 = 150 and 209 for C1 = 200 — the integral adds about 5 %, not the 17 % that would be needed to reach 234. That residual 12 % is precisely why the upper end of the band still leaves an 18.4 % error in location on the independent set, and it is the honest reason the band is offered as a bound rather than as a value.

dataset	n_points	Re_c_min	Re_c_max	mean_required	slope_per_decade_Re_c	correlation
Dagenhart & Saric (calibration)	6	1.92e+06	3.73e+06	153.1	-249.5	-0.901
Boltz et al. (independent)	4	9.50e+06	2.70e+07	233.8	32.1	0.58

Table 31. The required critical value against chord Reynolds number within each facility (crossflow_reynolds_trend.csv). The trends have opposite signs to the offset between the facilities.

11.5 Ablations — what each element of the formulation is worth

Each of the three elements that distinguish this formulation is switched off in turn, everything else held fixed, and the same two datasets the natural branch reaches are re-run. The table is regenerated by gen_validation.py and is not quoted from memory.

configuration	SS_err_pct	mean_abs_err_c	within_bracket	points	within_bracket_pct
no bubble closure	0.1	0.0629	19	86	22.1
one-equation laminar march	-10.2	0.0418	42	86	48.8
Drela-Giles envelope	-11.7	0.0431	45	86	52.3
full model	0.1	0.0384	50	86	58.1

Table 32. Ablation study (ablations.csv): Schubauer-Skramstad onset error and the 86 aerofoil conditions, one closure removed at a time.

Calibration. The solver is calibrated through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, clamped to the 0.0008-0.0298 range over which that correlation is quoted. The SAME constants reproduce every dataset here — five flat plates, two swept wings and 86 aerofoil conditions — demonstrating that no case-specific physics tuning is required, which is the essence of the universality claim.

Which sets are calibration and which are validation. The constants are frozen across every case, but three of them were SET on data in these tables, and a universality claim is only worth what its out-of-sample evidence is worth, so this is stated rather than left to be inferred. N_anchor, the units offset of the amplification scale, is set on the Schubauer & Skramstad plate. tu_hist, the weight on the flow history of Tu in the Abu-Ghannam & Shaw correlation, is set on the three ERCOFTAC T3 plates, which are the only bypass data here. CF_ratio, the cross-flow surrogate, is set on the Dagenhart & Saric wing. Everything else is out of sample: ERCOFTAC T3C4, which is the only test of the separation branch; all 86 NLF(1)-0416 conditions, which are the only test of the natural branch on a real aerofoil; and the Boltz et al. swept wing, which is the only independent test of the cross-flow branch. One constant per branch, each set on the smallest dataset that determines it, with the largest dataset for each branch held back.

11.6 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.

Validation	ERCOFTAC T3B flat plate	Roach & Brierley (1990), ERCOFTAC T3B (Tu=6%); Langtry & Menter, AIAA J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.
Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / Cf	Head M.R. (1958) entrainment method; Ludwieg H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Validation	ERCOFTAC T3A- flat plate	Roach & Brierley (1990), ERCOFTAC T3A- (Tu=0.87%); ERCOFTAC Classic Collection Case 020.
Validation	ERCOFTAC T3C4 flat plate (separation bubble)	Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020, variable-pressure-gradient series.
Validation	NLF(1)-0416 aerofoil transition (86 conditions)	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9; tunnel turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), NASA TP-2328, Fig. 25.
Validation	Swept-wing cross-flow (calibration set)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.
Validation	Swept-wing cross-flow (independent set)	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g).
Calibration	Linear stability / amplification database	Orr (1907); Sommerfeld (1908); Gaster M. (1962), JFM 14, 222; Mack L.M. (1977), AGARD CP-224; Drela M. & Giles M.B. (1987), AIAA J. 25(10), 1347.
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO

		2533:1975), FL360 properties.

Table 33. Validation, calibration and case-study sources.

dataset	source
Dagenhart & Saric 45 deg swept NLF(2)-0415 (cross-flow)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.

Table 34. Cross-flow calibration dataset provenance.

dataset	source
Boltz, Kenyon & Allen NACA 64(2)A015 untapered wing	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g); transition locations digitised by the present author as x/c = R_trans/R.

Table 35. Independent swept-wing dataset provenance.

dataset	source	n_points	Tu_pct	orifice_pitch
Somers NLF(1)-0416 aerofoil, Langley LTPT	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9 (transition) and Table I (coordinates); turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), 'Recent Modifications and Calibration of the Langley Low-Turbulence Pressure Tunnel', NASA TP-2328, Fig. 25.	86	0.03	0.05

Table 36. Aerofoil dataset provenance.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by putting all four on one scale of onset progress and firing at the first to complete — reproducing all regimes with one calibration set, and continuous in the free-stream turbulence across the natural/bypass handover.
- An amplification database in place of an envelope correlation: 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, tabulated against shape factor, momentum-thickness Reynolds number and frequency, with the march carrying one amplification factor per physical frequency. The stability solver reproduces the Blasius neutral point at Re_theta = 201 against the accepted 200.5.

- A separation-bubble closure that predicts a length rather than a point: the shear layer is carried across the dead-air region by the momentum integral with no wall stress — a step with no fitted constant, which reproduces the growth the T3C4 hot films record — and reattachment placed where the disturbance has amplified by the same N_crit used elsewhere, so the length scales with the disturbance environment.
- A two-equation laminar march whose closures are computed from the Falkner-Skan family rather than fitted, giving the shape factor a history. On the 86 aerofoil conditions it raises the number of predictions inside the experimental bracket from 42 to 50, improving both surfaces at once.
- Regime coverage with one constant set: transition-onset Re_theta_t predicted to +3.6 % on T3A, -16.5 % on T3A-, -7.3 % on T3B, -30.4 % on T3C4, +0.1 % on Schubauer & Skramstad, spanning 0.03-6 % free-stream turbulence intensity.
- An aerofoil validation built for this work: 86 transition locations digitised from Fig. 9 of NASA TP-1861 for the NLF(1)-0416 section, both surfaces, four chord Reynolds numbers and lift coefficients from -1.03 to +1.62, with nothing calibrated on them. Mean error 0.038 chord, and 50 of the 86 predictions fall inside the +/-0.025c bracket within which the experiment itself localises transition.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry → mesh → setup → solution → post-processing → validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle: 55 % laminar flow and 50 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated with a single calibration set against five flat plates, two independent swept-wing experiments and 86 aerofoil conditions, spans 0.03-6 % free-stream turbulence intensity. At cruise the wing achieves 55 % laminar flow and a 50 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver switches to the bypass route and predicts early transition, demonstrating regime coverage across the flight envelope.

Two limitations bound that claim and are stated here rather than left to be discovered. The cross-flow critical constant does not transfer between facilities: the two independent swept-wing experiments support the functional form of the criterion, and the measured data collapse on it, but reproducing the second requires a critical value 42 % larger than the first. And the method is an attached-flow formulation, so it has an incidence envelope — but that envelope is asymmetric and is set by the section, not by a round number. On the NLF(1)-0416 it declares its first condition at −9.5° of incidence, where the upper-surface layer separates within two per cent of chord, and it handles the same section down to −12.0° without complaint; at positive incidence nothing in the 86 conditions, which reach +8.7°, is declared, and a leading-edge bubble does not appear on the upper surface of the section itself until about +14°. Three of the 86 fall outside it and the method says so rather than returning a location. An earlier version of this report quoted ±8.5°, which was a plotting threshold rather than a measured envelope.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 113 (CSV: 52, PNG: 59, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV
01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG
01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG
01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG
03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_climb.npz	NPZ
04_solution/field_pressure_cruise.csv	CSV
04_solution/field_pressure_cruise.npz	NPZ
04_solution/integrated_forces.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/surface_climb_lower.csv	CSV
04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_length_sensitivity.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG

05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG
05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG
05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG
05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG
05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG
05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG
05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG
05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/ablations.csv	CSV
06_validation/aerofoil_nlf0416.csv	CSV
06_validation/aerofoil_nlf0416_source.csv	CSV
06_validation/aerofoil_nlf0416_summary.csv	CSV
06_validation/bubble_diagnostics.csv	CSV
06_validation/crossflow_criticals.csv	CSV
06_validation/crossflow_criticals_summary.csv	CSV
06_validation/crossflow_receptivity.csv	CSV
06_validation/crossflow_receptivity_summary.csv	CSV
06_validation/crossflow_reynolds_trend.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3AM.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/experiment_T3C4.csv	CSV
06_validation/residual_diagnostics.csv	CSV
06_validation/solver_SS.csv	CSV

06_validation/solver_T3A.csv	CSV
06_validation/solver_T3AM.csv	CSV
06_validation/solver_T3B.csv	CSV
06_validation/solver_T3C4.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/swept_wing_crossflow.csv	CSV
06_validation/swept_wing_independent.csv	CSV
06_validation/swept_wing_independent_source.csv	CSV
06_validation/swept_wing_source.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3AM.png	PNG
06_validation/plots/val_T3B.png	PNG
06_validation/plots/val_T3C4.png	PNG
06_validation/plots/val_aerofoil_nlf0416.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
06_validation/plots/val_swept_crossflow.png	PNG
06_validation/plots/val_swept_independent.png	PNG
07_equations/equations_index.csv	CSV

Table 37. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section c_l at cruise design incidence (alpha = 1.5 deg, M = 0.42)	0.517	-

Fig. 49. Fig. B1. geometry_definition.csv.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	1e-3 c
Max streamwise spacing	9.927	1e-3 c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wall-normal reconstruction stack (no volume mesh is generated)	-

Fig. 50. Fig. B2. mesh_metrics.csv.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Fig. 51. Fig. B3. material_properties.csv.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H*, l and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N, one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	C1 on Re_theta2, closed by an amplification integral
Intermittency closure	Narasimha universal (gamma)
Transition length	Dhawan & Narasimha, Re_lambda = 9 Re_x_t^0.75
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Laminar separation criterion	Thwaites lambda <= -0.09
Turbulent separation flag	H > 2.6
Drag integration	Squire-Young far-wake, evaluated at 0.98c
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	0.80 (reconstruction grid; no volume mesh is solved on it)

Fig. 52. Fig. B4. solver_settings.csv.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.4962	-
Section profile drag Cd	0.0045	-
Section Cd (counts)	45.0	counts
Section L/D	110.1	-
Section lift-curve slope a0 (panel)	8.111	1/rad
Zero-lift incidence a_L0 (panel)	-2.185	deg
Quarter-chord sweep	9.89	deg
Wing C_L (lifting line, taper + washout + sweep)	0.2548	-
Wing C_L / section c_l	0.513	-
Span efficiency e (lifting line)	0.8718	-
Wing induced drag C_Di (lifting line)	0.00272	-
Wing lift (lifting line)	23603.0	N
Wing C_L for level flight at MTOW	0.8998	-
Incidence for that C_L (lifting line)	7.55	deg
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6.414e+06	-
Mach number	0.42	-

Fig. 53. Fig. B5. integrated_forces.csv.